



Eos Family Expert Topics

Intro to Augment3d

Workbook

V3.0 Rev B

www.etcconnect.com/education

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Purpose of the Class

This class will provide an overview of the features of the Augment3D environment.

LEARNING OBJECTIVES:

After completing the class, one should be able to:

- Set up a native or tethered Augment3d environment.
- Use tools to navigate the Augment3d environment.
- Understand the principals of a digital 3D environment.
- Add and focus conventional, LED, and moving fixtures to Augment3d.
- Create, import, and modify 3D model objects.
- Understand file management for show data and model components.
- Use advanced focus and display tools within Augment3d.
- Understand and utilize XYZ focus palettes.
- Understand and utilize Staging mode.
- Create and use movable scenic channels.
- Understand the principles of Focus Position Estimation and Focus Wand.

WORKBOOK SYNTAX ANNOTATION

- **Bold** Browser menus
- **[Brackets]** Face panel buttons
- **{Braces}** Softkeys or buttons on touchscreen
- **<Angle brackets>** Optional keys or command line text
- **[Next] & [Last]** Keys to be pressed & held simultaneously
- **«Direct Select»** Direct Select button press
- **[MS Object]** Object on a Magic Sheet
- **Play Icon** Link to video on ETC's YouTube Channel – ETCVideoLibrary



HELP

Press and hold [Help] and press any key to see:

- the name of the key
- a description of what the key enables you to do
- syntax examples for using the key (if applicable)

As with hard keys, the "press and hold [Help]" action can be also used with softkeys and clickable buttons

THE MANUAL

The manual is available on the console, Tab #100.

Click on Add-a-Tab (the {+} sign) , select Manual

Hold [Tab] & press [100]

Please note that it is not available on Windows XP or Mac devices but is available as a download from the web site.

Introduction to Eos Augment3d

Augment3d is a suite of tools introduced in Eos 3.0, which adds a 3D programming and visual environment to the Eos Family of consoles, as well as ETCnomad for Mac and PC.

With Augment3d, you can control your rig directly on your console, or tethered on a graphics-centric computer. It allows you to focus lights with a click or a touch, preview your lighting looks in 3D space, and control your rig from your phone using augmented reality.

This class takes you through getting Augment3d operational, introduces you to 3D environments, and building and using your model and lighting rig.

AUGMENT3D PROGRAMMING ENVIRONMENT FEATURES:

- Add fixtures to the 3D environment in Patch, or by importing from a Vectorworks 2019, 2020, or 2021 file.
- Create architecture and scenery using stock objects from the Augment3d library or import models with support of over 50 file types.
- Set up graphics to fit your application, with many settings to balance quality and performance. Use visual assistance tools to identify fixtures and beams and set the ambient lighting to your choice of level and color.
- Speed up programming using tools like click-to-focus, preview mode, and blind edit – including effects.
- Attach objects or fixtures to desk channels, allowing you to move them for scene changes, or adjusting lighting positions throughout a performance or venue-to-venue.
- Use focus palettes to reverse-calculate fixture positions in the 3D environment, using a real lighting rig.
- Use augmented reality on your phone or tablet to focus lights, view rig-specific overlays on the camera, and view channel numbers at the source and the incident of the beam.

* Start with the following show file open:

"Eos Family Learning Series – Expert Topics – Rev A.esf3d"

Open on a console with a laptop connected via network and running Eos 3.0 in the shell (not launched)



Getting Augment3d Operational

The Augment3d environment can be launched in two ways – either in a tab, or on a Tethered machine. Running Augment3d as a tab allows low or medium graphics quality and is quick to open and use on any console. If you are using a large rig or prefer higher-quality graphics, running Augment3d on a tethered machine will allow the possibility for better speed and rendering.

OPENING AN AUGMENT3D TAB

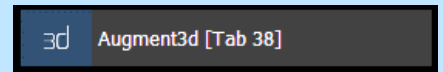
Consoles that have at least one Display Port as their monitor connections can run Augment3d in an internal tab.

OPEN AN AUGMENT3D TAB

On your non-CIA monitor, click (+) Add-A-Tab

Click on “Augment3d” (Tab 38)

Wait for the Augment3d space to initialize



Opens the Augment3d tab, wait for the Augment3d environment to initialize.

CLOSE AN AUGMENT3D TAB

Right mouse click on the Augment3d Tab (Tab 38)

Click on “Close Tab”

Closes the Augment3d tab

BEST PRACTICE IDEAS – RUNNING AUGMENT3D IN ETCNOMAD

- ETCnomad can run an Augment3d tab natively. The quality of the environment will depend on the computer being used. For more information, check the Eos 3.0 documentation for recommended hardware specifications.

SETTING UP AN AUGMENT3D TETHER

With an external computer connected, you can leverage the processing power of that device, which will allow Augment3d to run at higher performance.

You'll need a functioning network connection between the console and the tethered computer.

Note: the console and the laptop need to be running the same version of software.

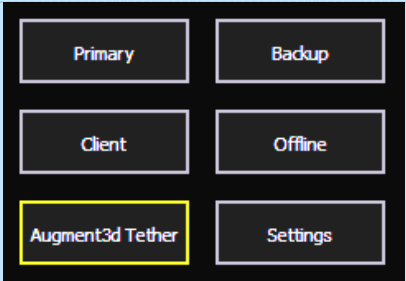
LAUNCHING AN AUGMENT3D TETHER

Make sure your console is booted up, logged in, and connected to the network

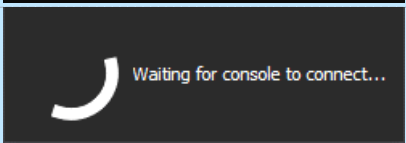
On your computer, launch Eos 3.0 software

If auto-launch is enabled, click on the "launching in 5..." countdown to override launching Eos.

Click on the button "Augment3d Tether"



Launches the Eos software in an Augment3d Tethered mode – you will see a dialog with "Waiting for console to connect..."



CONNECTING THE CONSOLE AND THE TETHERED COMPUTER

On your Console: [Displays] {Setup} {System}

Click on the {Augment3d} section on the left side of the screen

	Status	Name	Address	Version	Connect To Server
	Available	Augment3d Tab	Local		
	Available	Augment3d	10.101.92.9	1.0.0.432	

The window displays the local Augment3d tab, as well as all available tethers.

Click on the tether you want, and click "Connect to Server"

The console and the Tether are now connected.

Best Practice Ideas

- Only one instance of Augment3d can be open at one time. If you attempt to open an Augment3d tab on the console, when tethered, it will display a warning that you are tethered.

AUGMENT3D UNITS

Click on the Metric/Imperial Dropdown Menu

Meters – typing [1.7] will result in 1.7 meters

Feet – typing [1.7] will result in 1.7 feet (1'-8.4")

Feet/Inches – typing [1.7] will result in 1 ft, 7 inches (1'-7")

Select {Feet/Inches}

Metric/Imperial

Feet/Inches

DEVICE ENABLE A3D

This option allows you to enable or disable the ability to open an Augment3d tab on a per-device basis. It is Enabled by default.

Device Enable A3D

Enabled

USER ENABLE A3D

This option allows you to enable or disable the ability to open an Augment3d tab on a per-User basis. It is Enabled by default.

Click on the {Users} section on the left side of the screen

Cue Settings	User ID	User	Online	Augment3d	Name
Show Control	1	1	Online	Enabled	
Output					
Remotes					
Partitions					
Users					
Augment3d					

AUGMENT3D CONNECTION STATUS

Once connected to a tether, "Tether" appears in the network status of the console. Press [Live].

User 1 | Primary | Master | Tether

BEST PRACTICE IDEAS

- The quality of the model environment will depend on the computer being used. For more information, check the Eos 3.0 documentation for recommended hardware specifications.
- The tethered computer is still a separate machine – you will need access to that device's mouse and keyboard to navigate and edit the Augment3d environment.
- You do not need an ETCnomad dongle to run a computer in Tethered mode.
- You can learn Metric/Imperial selection into a Macro.



Basic Displays & Navigation

Make sure you have "Eos Family Learning Series – Expert Topics – Rev A.esf3d" loaded on your console.

GETTING STARTED WITH CHANNELS

The Augment3d display is a Live/Blind view into the system, much like Tab 1 and Magic Sheets. Even in a Tethered environment, the Augment3d display will follow and affect the mode of the desk.

[Live]	
[101] [Thru] [105] [At] [Full] [Enter]	Check out the Augment3d display
Tilt and pan the units.	Watch the Augment3d display change with the manual data changes
Make them blue.	
Drop in a Gobo.	
Form – zoom out a bit, sharpen the gobo	
[Clear] [Sneak] [Enter]	
Hit [▶] (Go) a few times	Watch the display change
[1] [Thru] [5] [At] [50] [Enter]	Select conventionals
[Effect] [412] [Enter]	Run the effect, watch it run
[Sneak] [0] [Enter]	Stop the effect
[Blind]	Augment3d now goes into a Blind mode, just like a channel display
[Next] , [Next], [Next] , [Next], [Next] , [Next], [Last], [Last],...	Augment3d shows what is stored in each cue
[Cue] [37.2] [Enter]	Look at the effect running in Blind
[Live]	Back to the Live state of the desk



NAVIGATING THE AUGMENT3D ENVIRONMENT

You can use a mouse and keyboard to navigate the 3D space of Augment3d.

From a Front View, the axis labels are in the following orientation:

X = Red – Left & Right movement

Y = Green – Nearer & Farther movement

Z = Blue – Up & Down movement (altitude)

KEYBOARD NAVIGATION

On the tethered machine keyboard: Left & Right arrows

Moves the camera left and right

Up & Down arrows

Moves the camera up and down

[Shift] & [Up & Down Arrows]

Dolly the camera in and out

With Augment3d open in a tab on a console, you can use the Page Arrows and Shift to navigate when the tab is in focus.

MOUSE NAVIGATION

Left Mouse click will select fixtures (we will look at this later).

Right mouse button click & Hold

Look around

Roll mouse wheel up and down

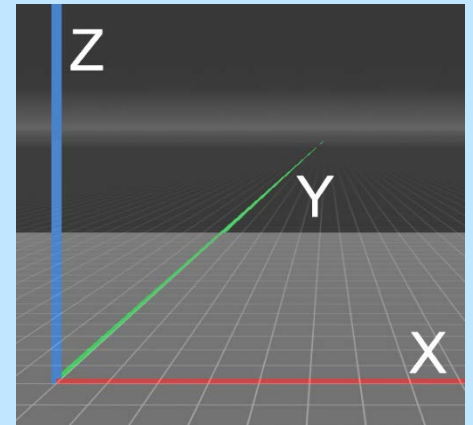
Dolly the camera in and out

Hold mouse wheel down while moving the mouse

Moves the camera up, down, left, right

Try looking in a different direction, and rolling your wheel

The camera dollies in and out along the direction you are looking



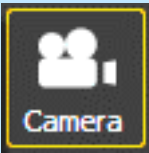


CAMERA MENU

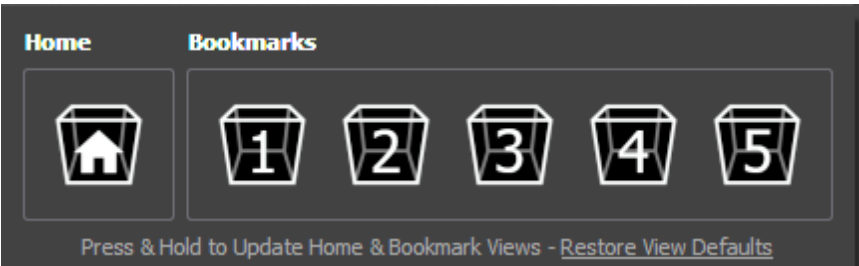
The Camera Menu allow you access to quick views, as well as other camera settings.

CAMERA MENU

Click on the {Camera} menu on the Augment3d Toolbar



QUICK CAMERA VIEWS



At the top of the Camera Menu are quick camera views. All views can be customized and restored to default.

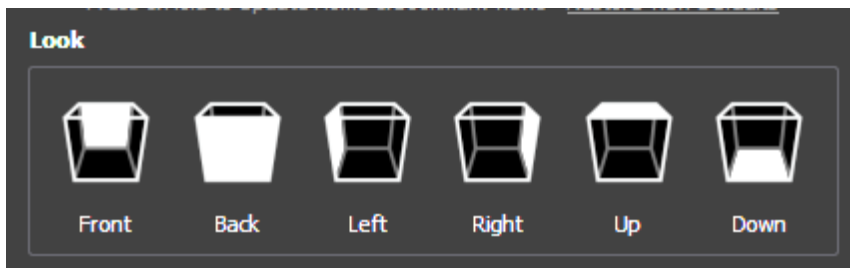
Click on {Home}	Moves the camera to the default home position
Click on {Bookmark 1}	Moves the camera to Bookmark 1
Click off the menu, Right Mouse Button Click & Hold	Move your camera to a different location
Click on the {Camera} menu Left click & hold on {Bookmark 1} until "View Updated" appears	Saves your new camera view to Bookmark 1
Click on {Home}, then on {Bookmark 1}	Your new camera location is now easy to access.
Click on {Restore View Defaults}	Restores views to Eos default settings

You can change the home bookmark using this same method.

BEST PRACTICE IDEAS

- For command line and macro accessible camera views, use Snapshots.

QUICK LOOK VIEWS



In the center of the Camera Menu are quick look views. These allow your camera position to remain stationary in space, but for you to look in a different direction.

Click on {Back}

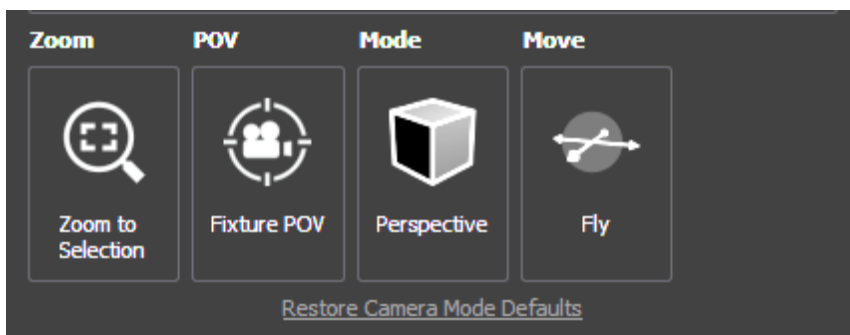
Turns the camera to look back

Click on {Up}

Turns the camera to look up

Click on the {Home} camera view

Restores the camera location and view



At the bottom of the Camera Menu are other camera settings.

ZOOM TO SELECTION

Type [101] [Enter]

Click on {Zoom to Selection}

Moves the camera to see Channel 101

FIXTURE POINT-OF-VIEW

Click {Fixture POV}

Camera moves to within gate of chan 101

[102] [Enter], [Full] [Full] then move Tilt, and Pan

Camera shows what the fixture is doing

{Fixture POV} to turn it off

The camera returns to its location before Fixture POV was enabled.

CAMERA MODE

In **Perspective** view (default), objects which are far away appear smaller than those nearby. Perspective viewpoints give more information about depth and are often easier to view because you use perspective views in real life.

In **Orthographic** view, all objects appear at the same scale. Orthographic viewpoints make it much easier to compare two parts of a model, as there is no question about how the viewpoint may affect the perception of distance.



Click on {Perspective}, changing it to Orthographic

Right Mouse Click & Hold to look around

Notice the difference in the feel of the model

Click on {Camera}, {Orthographic} changing it to back to Perspective

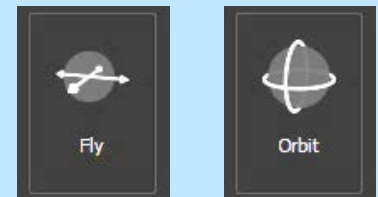
Restores the mode to Perspective

CAMERA MOVE

Camera Move settings affect how your Look function operates (right click & hold).

In **Fly** view (default), the Look function will keep the camera stationary in space and turn it to change the view. This is similar to how many video games function, like Minecraft.

In **Orbit** view, the Look function will orbit the camera around a pivot point, moving the camera spherically, changing both view and camera location in space. This is how many 3D drafting tools operate, like Vectorworks.



Click off the menu to close it, use your look function (Right click & hold)

In Fly mode, camera is stationary, but rotates to change your look

Click on {Camera} menu, then {Move}, changing to Orbit

Click off the menu to close it, use your look function (Right click & hold)

In Orbit, the camera moves around the selection, pivoting the camera to change the look

Click on {Camera} menu, click {Orbit} to change back to Fly mode

Press & hold ALT and use your Look function

Temporarily toggles between Fly and Orbit. Releasing ALT restores to what is set in the Camera Menu

Anytime you move your camera, Orbit will reset to the new location.

BEST PRACTICE IDEAS

- Snapshots will remember Fixture POV, Mode, and Move settings – so you can easily recall these changes without having access a mouse.



Display Quality Settings

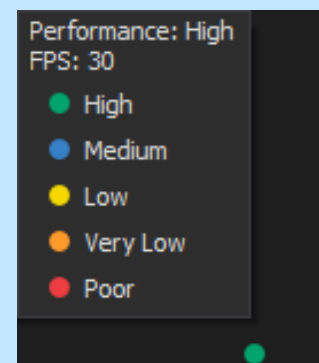
The quality of the Augment3d environment will depend on the computer being used. You will get different quality depending on whether you are using an Eos Family console or a tethered laptop. To balance quality of graphics with speedy performance, you can adjust the Display Quality Settings.

In the bottom right of the Augment3d window, there is a dot that will change color based on the current quality of the graphic performance. The metric is displayed in Frames Per Second (FPS).

PERFORMANCE POPUP

Hover over the dot with your mouse to see the Performance popup:

- High: 25-30 FPS
- Medium: 20-24 FPS
- Low: 15-19 FPS
- Very Low: 10-14 FPS
- Poor: 0-9 FPS

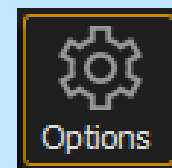
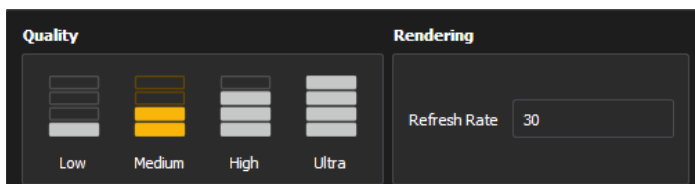


QUALITY & RENDERING

Quality determines to what level of detail the software will render the Augment3d model. Rendering determines how fast the software will refresh the model.

To adjust the quality settings, click on the {Options} menu and then {Quality}

- **Quality** – Low and Medium are available on consoles.
 - In addition to Low and Medium, High and Ultra are available on ETCnomad and Tethered devices.
- **Rendering** – this sets the maximum Frames Per Second the software will attempt to achieve.



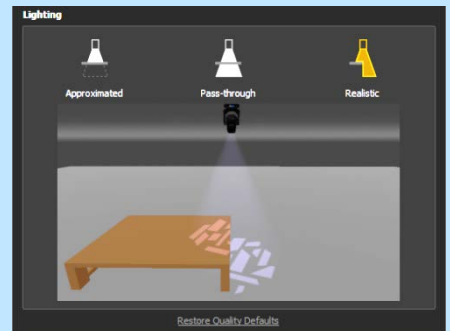
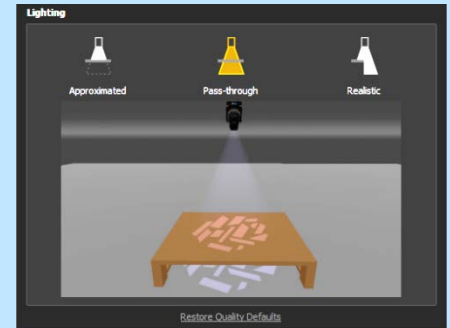
LIGHTING

Lighting determines how beams intersect with surfaces, and whether they cast shadows.

- **Optimized (Pass-Through)** – beams will appear on all objects that the beam intersects, including those behind other surfaces. And will not cast shadows
- **Approximated** – beams will end at the first object they intersect, and not continue.
- **Realistic** – beams will appear on all objects that they intersect with and will cast shadows.

BEST PRACTICE IDEAS

- The default Quality settings are Medium & Approximated. You can adjust the settings to meet the performance level you want, based on the capabilities of the hardware you are running.
- Eos Family Consoles only have access to Low and Medium Quality settings.





Starting A New Show File

Augment3d handles fixtures and model components separately. This allows you to keep your venue loaded in Augment3d, make changes to scenic objects, and remove your fixtures as you start a new show.

[Displays] {File} {New} [Enter]

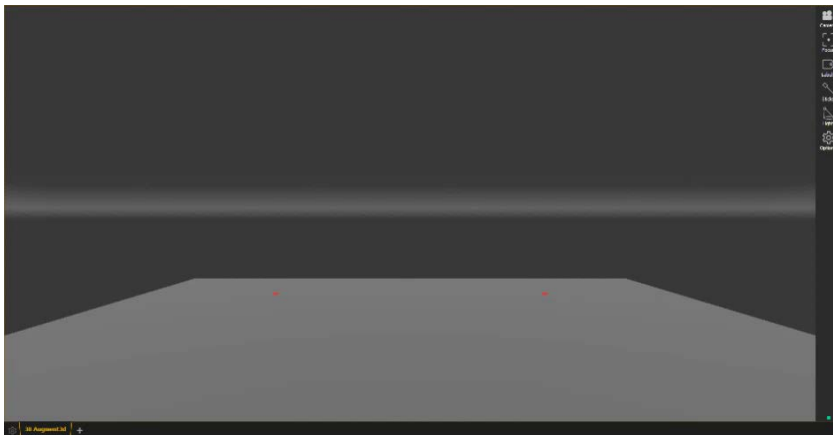
Click on {Continue without Saving}

If Preserve A3D is unchecked, the model will be purged when starting a new show file.

If Preserve A3D is checked, the model will remain when starting a new show file.

Uncheck "Preserve A3D Scene," and click OK

You will have a completely blank show file, and a blank Augment3d space. A blank Augment3d window will appear. A floor is placed in the model by default.

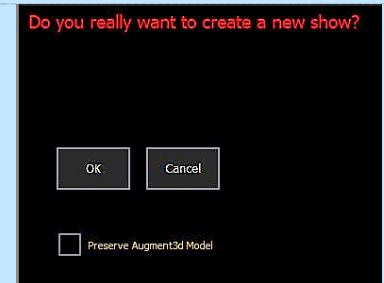


BEST PRACTICE IDEAS

- When creating elements to import into Augment3d, it is a good idea to have separate files for your venue and your scenery. At the end of a show, you can start a new show file with "Preserve A3D" enabled. It will keep the venue and scenery in the new show file. You can then delete the old scenery and import new scenery for the new show.

Get ready to start a new show file

Don't want to record over the Expert Topics file





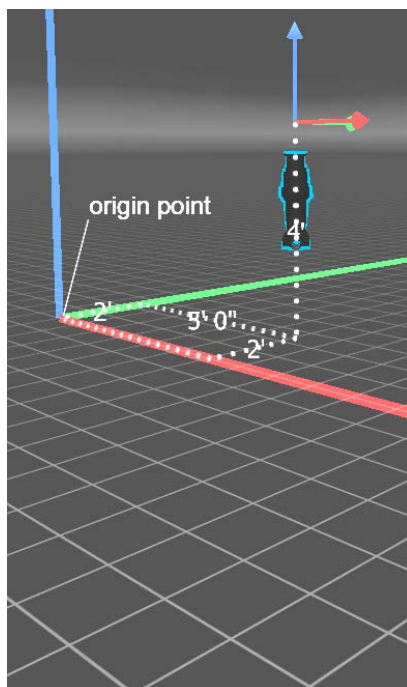
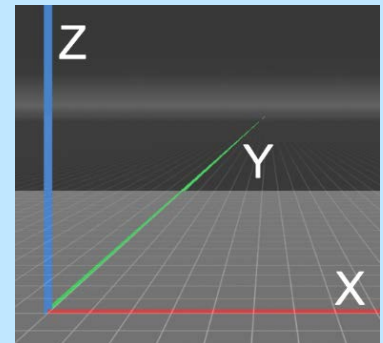
About 3D Environments

ABOUT ORIGIN POINTS

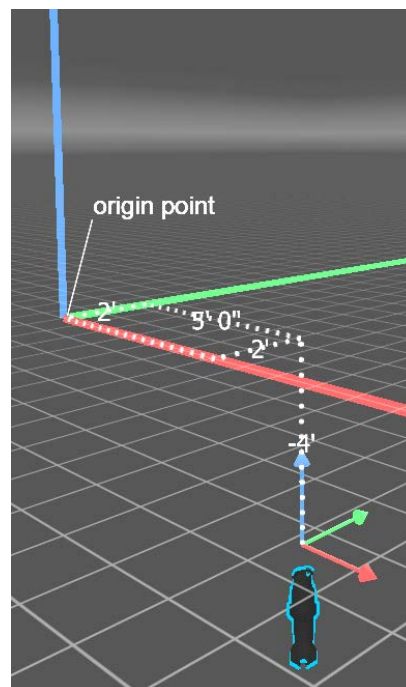
In a three-dimensional model space, we need to be able to define an object's location in reference to some other known place. To serve this purpose, model spaces have an **Origin Point**, which is a fixed location in space. The Origin is so called, because it is the origin point for the X, Y, and Z axes. At the Origin Point, the values of all axes are zero.

To describe an object's **Position** in space, we can define the offset from the origin for each axis. If we define X at 5'-0", Y at 2'-0", and Z at 4'-0", the object goes to that position in space. Each of these offsets can be negative as well, getting the object on the other side of the origin.

In a typical proscenium theater, your x-axis might describe your stage left and stage right movement, your y-axis might describe your upstage and downstage movement, and your z-axis might describe your height above or below the floor,



An object with defined X, Y, and Z coordinates, with Z having a positive value (4'-0").



An object with defined X, Y, and Z coordinates, with Z having a negative value (-4'-0").

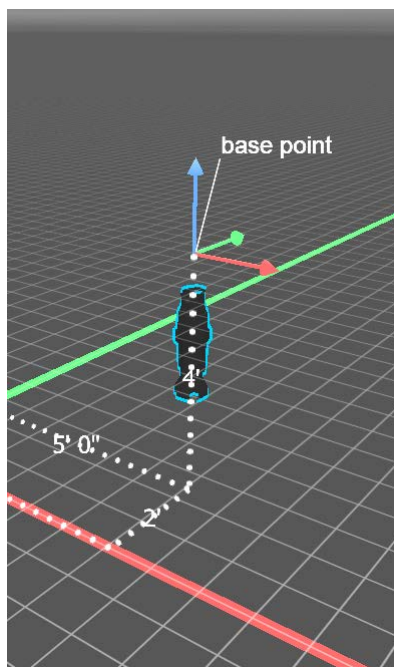
ABOUT OBJECT POSE

Spatial position isn't the only important information about an object, we also want to know its **Rotation** along each axis.

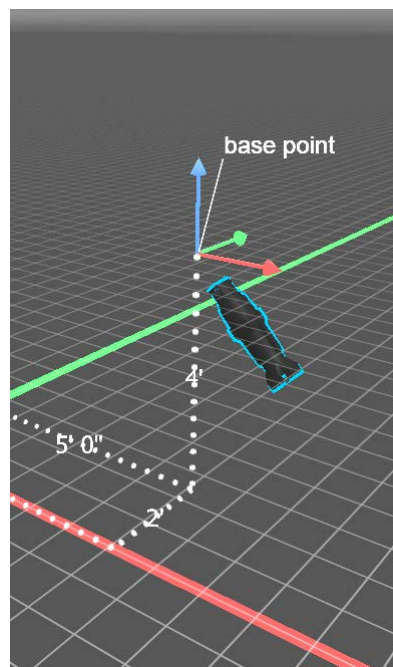
That means there are a total of six pieces of information that define the position *and* orientation of an object in space. This collection of data is often referred to as the object's **Pose** information.

ABOUT OBJECT BASE POINT

All objects have a **Base Point**, which is where on the object the **Position** and **Rotation** values are referenced to. Many drafting programs allow you to define an object's Base Point.



An object with a Base Point where the c-clamp would be located.



Giving the Y Rotation field a value of -45 degrees rotates the object on the Y axis at the base point.

BEST PRACTICE IDEAS

- Setting your Origin to be at an easily-referenceable location will help you build and interact with your model more quickly (for "Mind the Gap," we set the Origin to X = Center Stage, Y = Upstage Plaster Line, Z = Stage Floor).

Adding Static Fixtures

When adding conventional or static fixtures, you add them to your Augment3d Model simply by defining their Position and Orientation in Patch. These values should match their real-life distances from the Origin Point.



ADDING CONVENTIONAL FIXTURES

[Patch] [11] [Thru] [15] [At] [11] [Enter]

Patches channels, defaults them as a Dimmer type. The fixtures have not yet been added to the Augment3d Model

[11] [Enter]

Selects only channel 11

Click on the {Augment3d} section in Patch

In the Position & Orientation tab: click/touch on the "Z" Feet/Inches field

Don't forget that your measurement type is set in System Settings > Augment3d

The screenshot shows the 'Patch' software interface. On the left is a sidebar with 'Patch', 'Attributes', 'Database', 'Augment3d' (highlighted with a yellow arrow), and 'Device List'. The main area has a 'Position & Orientation' tab selected, with 'FPE' as a sub-tab. Below this, there are three columns for X, Y, and Z. Each column has a 'Feet/Inches' row and a 'Degrees' row. In the 'Feet/Inches' row, the Z field is highlighted with a yellow box and contains '0''. Below the 'Feet/Inches' row are rotation controls (curved arrows) for each axis.

Type or input [10.6] [Enter]

Notice the command line: X/Y/Z. The Z value is set to 10'-6", and the fixture appears in the Augment3d Model

Click/touch on the "X" Degrees field

This screenshot is similar to the previous one, but now the 'X' field in the 'Degrees' row is highlighted with a yellow box and contains the value '0'. The 'Z' field in the 'Feet/Inches' row now displays '10' 6"'. The rotation controls remain visible at the bottom.

Type or input [45] [Enter]

The fixture is rotated 45 degrees upstage along the X axis. You can use negative values

[Clear] to clear the command line

Notice fixtures are lit up when selected, and will go back to their Live state (in this case, off) when deselected.

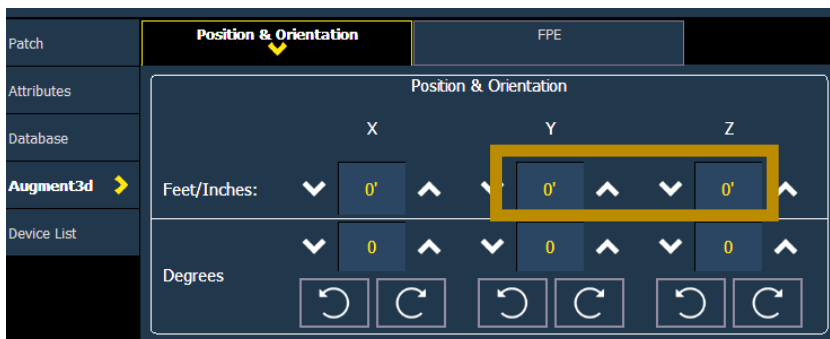
SETTING FIXTURES' POSITIONS INDIVIDUALLY

[12] [Thru] [15] [Enter]

Selects channels 12 through 15

Change the Values for Y to -5'-0" (negative 5 ft), and the Values for Z to 10'-0".

All four fixtures will move to the same location in space. They appear like a single fixture, because they are on top of one another.



Now spread them out along the X axis:

[Next] {x} [-] [5] [Enter]

[Next] {x} [-] [2.6] [Enter]

[Next] {x} [2.6] [Enter]

[Next] {x} [5] [Enter]

The fixtures will end up in a line at -5'-0", -2'-6", 2'-6", and 5'-0".

SETTING FIXTURES' POSITIONS IN A RANGE

[11 thru 15] [Enter]

{Position} [-] [10] [Thru] [10] [Enter]

Watch the command line. Fixtures spread out along the X-axis.

The command line displays the coordinates in an x/y/z order (-10/*/* Thru 10/*/*).

With channels still selected, {Y} [-] [5], {Z} [10]

To bring Channel 11 in line with channels 12 thru 15

Under X Degrees, bring Rotation back to 0

To bring focus in line with other channels

{Position} [-] [5] [/] [/] [7] [Thru] [5] [/] [/] [10] [Enter]

Spreads across pipe and at different heights

OFFSETS FROM CURRENT POSITIONS

{Position} [/] [/] [+] [2] [Enter]

Fixtures all move up to 2 feet from last position.

{Position} [/] [/] [+] [-] [2] [Enter]

Resets them back to previous position.

A [+] always indicates an offset, with a [+] [-] indicating a negative offset. In this example, without the [+], the console would put all the fixtures at the absolute measurement of -2'-0" from the origin.



ADDING STATIC LED FIXTURES

Non-moving LED units can be added to the Augment3d Model in the same way as conventional units.

While still in Patch

[51] [Thru] [53] [Enter]

Selects channels 51 through 53

**In the {Patch} section, change the Type to {ETC Fixtures}
{ColorSource Spot [1] +3D}, {Direct [6] +3D}**

Sets Type to fixed-focus LED units. (The "+3D" means Augment3d compatible).

[At] [2] [/] [1] [Enter]

To complete the Patch

Click on the {Augment3d} section

**Click or Touch the {Position} tile in the Properties area, then
[/] [2] [/] [35] [Enter]**

Puts {Position} on the command line, then allows you to enter X/Y/Z position data. Notice no X position yet. All fixtures go to the new Y position.

Now Place them along the X axis:

[Next] {Position} [24] [Enter]

[Next] {Position} [16] [Enter]

[Next] {Position} [4] [Enter]

Changes just the X property, leaving Y and Z as they were. Notice assumes X value without additional slashes. To get to just Y, use [/], and to get to just Z, use [/] [/]

[Select Last] [Enter]

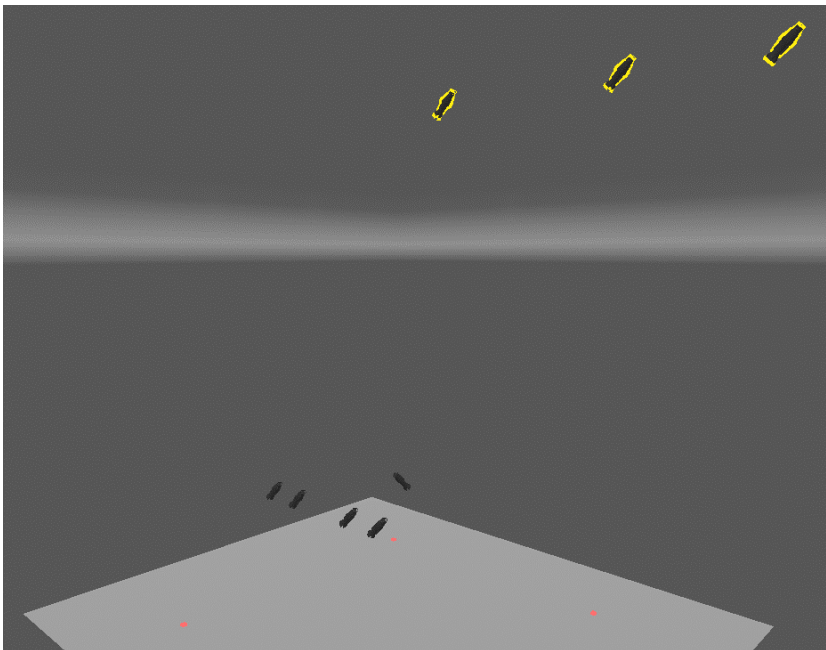
{Y Orientation} [45] [Enter]

Rotates the fixtures on the Y orientation, as they are pipe-end units.

In your Augment3d Window, zoom to see the fixtures:

This is a good way to rough in the rotation of the fixtures. We will cover using click-to-focus in the following chapter.

This is a good way to rough in the rotation of the fixtures. We will cover using Click-To-Ffocus in the following chapter.





Focusing Static Fixtures

Static fixtures can be focused in the Augment3d model using Click-To-Focus. Select them in Patch and then click on them in the model.

It is best to have 2 monitors or 2 screens to do this. If you only have one monitor, you can set up a split screen with the Patch and Augment3d tabs.

In the Augment3d model, get a camera view similar to the one on the previous page.

Still in Patch, [51] [Enter]

Channel 51 turns on in Augment3d (not on stage).

The Augment3d model displays anything that is being output live in the rig as well as anything that is selected in Patch. This allows you to make changes and see how they are going to affect your model without turning on the fixtures in the live rig.

To focus the selected fixture, find a place on the floor you want it pointed

You must have an object to focus on – in this case, the floor plane.

In the model, click & hold on a location. Let go of the mouse button once the fixture moves.

Channel 51 will aim itself at the location. Notice the changed Orientation data in Patch.

Find another location. Click & hold the mouse button and then drag to make small adjustments.

The fixture will follow your mouse pointer until you release the button.

[Clear] to clear the command line

Channel 51 turns off in the Augment3d model.

Still in Patch, [51] [Thru] [53] [Enter]

All selected channels turn on in Augment3d (not on stage).

Back in the model, click & hold on the location. Let go of the mouse button once the fixtures moves.

All three channels will aim themselves at the location.

Find another location. Click & hold the mouse button after the fixture moves, and drag the mouse.

The fixtures will follow your mouse pointer until you release the button.

{Orientation} [0] [/] [45] [/] [0] [Enter]

Similar to {Position}, same X/Y/Z syntax

[Clear]

BEST PRACTICE IDEAS

- If you are using Augment3d in a tab on a single monitor, it is easiest to use a split screen, and to have Patch in one frame, and Augment3d in the other frame.
- Because you must have an object to click on, for something like a side light system, it may be useful to have a vertical plane object at the center line for focus, that can be hidden later.



Additional Fixture Properties

Static fixtures can store and utilize more than just Pose (position and rotations) properties. You can store beam angle, beam edge, shutters, gobo image & rotate, gel, and more.

BEAM ANGLE

Still in Patch, [51] [Thru] [53] [Enter], in the {Augment3d} section

{Beam Angle} [26] [Enter]

Changes the fixture beam angle to 26 degrees.

Look at the Augment3d window

The beam angles match the lenses in the real unit.

BEAM EDGE

Still in Patch, [51] [Thru] [53] [Enter], in the {Augment3d} section

{Beam Edge} [75] [Enter], look at the Augment3d window

The Beam goes to a soft edge. For beam edge, 50 is always sharp, regardless of throw distance. 0 and 100 are soft.

{Beam Edge} [50] [Enter], look at the Augment3d window

The beam goes to a sharp edge.

{Beam Edge} [Enter], look at the Augment3d window

With no value, the beam edge is set to sharp.

SHUTTERS

Still in Patch, [51] [Thru] [53] [Enter], in the {Augment3d} section

{Shutters} make adjustments

You can adjust the angle and thrust of the conventional shutters. Because the Augment3d tab is a Live/Blind view, you will need to go back into live to check your work.

[Select Last] {Shutters} [Enter]

Clears shutters from the fixture.

GOBO

Go to the {Database} section

[Select Last] {Gobo}

Opens the Gobo Library



Manufacturer & Library on the Left, Gobos on the Right

Choose your favorite breakup. In video, selects Rosco - Steel Gobos - Lines2

GOBO ROTATE

Back in the {Augment3d} section, {Gobo Rotate} [90] [Enter]

Rotates the gobo in its gate, so you can align a gobo correctly (like a window gobo).

GEL

[11] [thru] [15] [Enter] , go in to {Database} section

Adds Roscolux R80 to the unit.

{Gel} on the keyboard type [R80] [Enter]

HIDE BEAM

[15], go in to {Augment3d} section, select {Hide Beam}

Enables Hide Beam

Int the model, Go to the Camera menu

Click {Zoom to Selection} and {Up}

To move the camera to see front of fixture

Can see light in the fixture, but there is no beam in the model.

[Clear]

Clears the selection

The FPE Tab and FPE Set field are discussed in the Fixture Position Estimation section.

BEST PRACTICE IDEAS

- If you are using an Augment3d tab, it is useful to use a split screen with the Augment3d tab in one frame, and the Patch tab in the other frame.
- While in Patch, the Augment3d tab will continue to show the contribution of the Live state. When channels are selected, they turn on to full.
- You may not need all your static fixtures in your model if they are not adding value to your design process. Sometimes adding static fixtures can be more time-consuming than it is worth. If you just need the status of your static fixtures in a visual format, consider a Magic Sheet with channel objects linked to color and intensity.



Adding Moving Lights

Moving fixtures are patched and positioned in the same way as static fixtures, but because most of their attributes are provided by the consoles, they need less information in Patch.

ADDING MOVERS

In the Augment3d display, click {Camera} {Home}

In Patch, [101] [Thru] [105] [Enter]

Selects channels 101 through 105

**{Type} {High End Systems} {SolaFrame Theater [47] +3D},
{SolaFrame Theater [47] +3D}**

Patch the channels as movers.

[At] [2] [/] [201] {Offset} [50] [Enter]

Click on the {Augment3d} section

Click or touch the {Position} tile in the Properties area

Puts {Position} on the command line

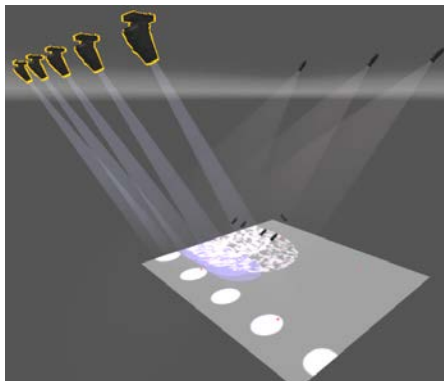
[-] [19] [/] [-] [29] [/] [34] [Thru] [19] [/] [-] [29] [/] [34] [Enter]

Allows you to enter X/Y/Z position information, spreading the fixtures across a pipe (X -19 to 19) located in the FOH (Y -29), hung 34' in the air (Z)..

[Live] [Select Last] [Full] [Full]

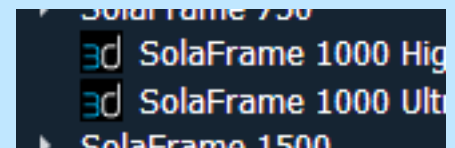
Turn on the fixtures in Live, will see them on stage and in the Augment3d model.

Tilt up, zoom out a bit, add a Gobo, and then add color



BEST PRACTICE IDEAS

- Fixtures that don't have model data use a generic model of whatever type of fixture the profile categorizes it as (wash, spot, ellipsoidal, etc.). At 3.0 release, there will be over 60 unique fixtures with proper models, and more will be added over time. Fixtures with verified models have the Augment3d badge next to their profile name in Patch > Type > Search.
- Fixture Position Estimation (FPE) is a great way to quickly learn the Poses of your moving lights from a real rig. Check out the FPE section to learn more.





Removing Fixtures

Channels can be removed from the Augment3d Model, but remain in the show file, simply by removing their Position data.

REMOVING AN ADDRESS FROM A CHANNEL

In [Patch]

[101] [At] [Enter] [Enter]

Removes address information from the channel, but maintains the fixture in the Augment3d model space.

REMOVING A FIXTURE FROM AUGMENT3D ONLY

In [Patch], go to the {Augment3d} section

[102] [Position] [Enter]

Removes only position and orientation data from the channel. The channel remains patched to its addresses, and can still store and play back data – it is just no longer in the Augment3d model space.

Note that doing {Orientation} [Enter] will only set all orientation values to zero.

UNPATCH A FIXTURE

[103] {Unpatch} [Enter]

Removes the patch, pose, and all non-intensity show data.

DELETING A FIXTURE FROM THE SHOW

[Delete] [104] [Enter]

Deletes channel from the entire show, including Augment3d data.

[Live], look at the Augment3d window

Channel 104 is no longer in the Augment3d display, and has been fully removed from the rest of the show file.

BEST PRACTICE IDEAS

- If you are using fixtures on a real rig that you no longer need represented in your Augment3d model, removing the Position data keeps show data preserved, and simply hides the unit in the Augment3d model.
- Be careful about removing Position data if you are using XYZ Focus Palettes.



Intro to Edit Mode

Like other areas of the Eos software, the Augment3d suite has an editor which allows you access to advanced configuration and usage features.

OPENING EDIT MODE

On the Augment3d device, click on the {Edit} button.

LEAVING EDIT MODE

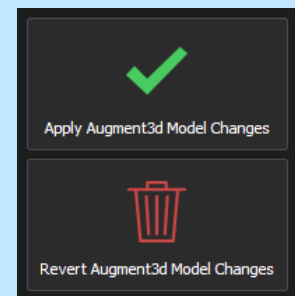
To leave Edit Mode, click on the {Done} button.

You will be prompted to either save or discard the changes you have made since entering Edit Mode:

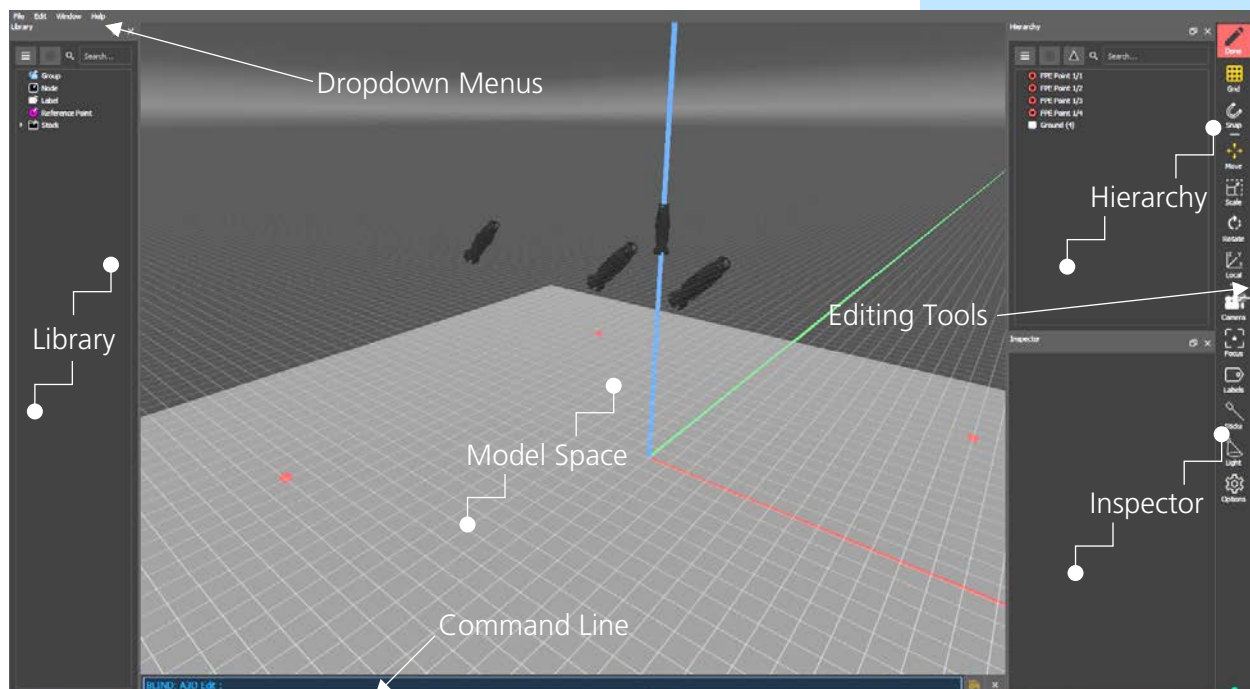
- **Apply Augment3d Model Changes:** this will save any edits you have made since entering Edit Mode. In a multi-console system, these changes will be sent to other devices only after Apply Changes has been clicked.
- **Revert Augment3d Model Changes:** any changes that have been made since entering Edit Mode will be discarded.

Because fixture data is stored in Patch, any changes made to fixtures while in Edit Mode will be stored. Apply or Revert changes pertains to non-fixture assets only.

For now, click off this menu



EDIT MODE OVERVIEW



MODEL SPACE

The model space is where you make changes to the Augment3d Model. All navigation tools work the same in Edit Mode as they do in Control mode.

COMMAND LINE - BOTTOM

The command line appears at the bottom of the Model Space. There is a button on Tethered devices that allows local keyboard input to be sent to the console as command line functions.

LIBRARY – ON LEFT

The library contains a built-in collection of scenic elements, shapes, and drawing tools. The library is also where items that are imported into Augment3d are stored and can be brought into model space.

If Library is not open when you hit Edit, go to the upper left dropdown menus, click on Window, then Library. If you leave open when you hit done, next time you edit, it will be open.

HIERARCHY – ON RIGHT

The Hierarchy contains all objects in the Augment3d Model space, which are called **Assets**. This includes model objects contributed from the library, fixtures contributed from Patch, and FPE/Reference Points.

INSPECTOR – ON RIGHT

The Inspector allows you to change properties for any items you have selected in your Hierarchy.

DROPDOWN MENUS – TOP LEFT

The dropdown menu has access to mouse-driven cut/copy/paste functions, and other menu visibilities. There is also access to Saving and Resetting the Default Model. In a Tethered mode, this is where items are imported into Augment3d.

WINDOW

If you click on {Window} in the dropdown menus, you can access which menus are shown, including some additional items that are not displayed by default.

Preferences

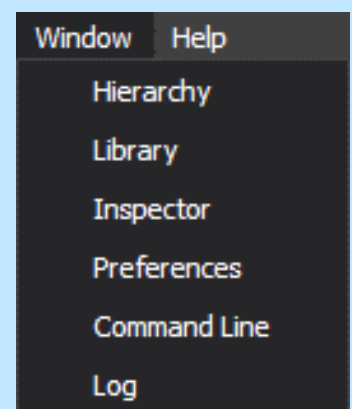
The Preferences dock gives quick access to settings and offers some settings that are not available in Model Tools.

Log

The log displays troubleshooting information, if you are experiencing issues with the Augment3d environment. (Diagnostics)

All these menus are draggable, dockable, and resizable.

For now, close the Preferences and Log menus if opened





EDIT MODE – GRID/SNAP

When moving or rotating objects, a Snap Grid assists with drawing and placing objects precisely.

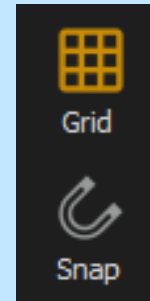
GRID

The Augment3d environment can display a snap grid for reference. It displays perpendicular to the Z axis. You can adjust the granularity of the grid in the Preferences Menu by changing the Snap distance, without enabling Snap.

In Edit mode, Click on {Grid} directly under the Edit/Done icon

Click on {Window} in the dropdown menu, click {Preferences}

Under Snap Distance, input 6" [Enter]



This turns on the Grid in the model

The grid is now displaying with lines every 6 inches.

SNAP

With Snap enabled, objects that are being dragged will lock to the grid intersection points.

Click on {Snap}

To enable the feature

Even if the Grid is turned off (click {Grid}) to avoid visual clutter, Snap will still function if enabled.

Click on {Snap} again

to disable the feature, objects can be dragged freely



EDIT MODE – MOVE/SCALE/ROTATE

The Editor provides tools that allow mouse-driven modifications for the objects in your model.

MOVE

The Move tool allows you to select objects, and move them along the X, Y, and Z axes.

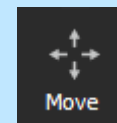
Click on the {Camera} menu, then the {Home} bookmark

Click on the {Move} icon to select it

Click on a fixture, to see the Move tools for that object.

Click and drag on each of the three axes arrows to move the fixture around.

As you move the object, notice the X, Y, and Z data changing in the Inspector window.



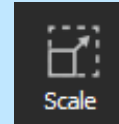
The fixture turns blue, indicating it is selected in the Augment3d Editor.

Notice the measurement references to the Origin. It also snaps to the 6" distance of the grid.

This is connected to the X, Y, and Z information in Patch, and is being updated real-time in both places

SCALE

Scale allows scenic objects to be adjusted to larger or smaller than they currently are, along each axis. Scale does not work on fixtures.



Click on the floor plane in the model - zoom back/out to see it

Click on the {Scale} icon to select it

Click and drag the green or red nodes to stretch the object on the x or y axis.

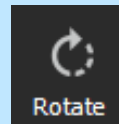
The Z handle (blue) will not function because the object is a 2D plane.

Click on the white node to stretch proportionately in all directions

Including Z if available

ROTATE

Rotate will orbit an object around each axis, allowing it to stay in the same position, but to rotate about its base point.



{Camera}, {Home}

Select a fixture

Click on {Rotate} so that it is enabled

Notice the “planet rings” around the object

Click and drag on any color of ring, rotating the object along the corresponding axis.

Notice it is rotating the object on the base point, not necessarily on the center of the object.

Rotate with Snap

With Snap enabled, rotate will lock to predefined degrees. The default is 5 degrees but can be changed in Preferences.

{Preferences}, Angle, and change it to 15, and hit [Enter].

Click on {Snap}

To enable

Rotate the fixture

Noticing it snaps to every 15 degrees

Click on {Snap} again

To toggle off the Snap feature

Rotate the fixture

Notice it rotates freely again

BEST PRACTICE IDEAS

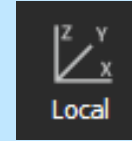
You cannot add fixtures in the Augment3d Editor, you must patch the units in the Patch tab, and give them basic Pose data. You can then enter the Augment3d Editor and make changes to the Pose data with the tools in the editor.



EDIT MODE - GLOBAL & LOCAL ORIGIN

In a 3D model space we need to be able to define an object's location in reference to some other known place. To serve this purpose, model spaces have an origin point which is a fixed location in space. The origin is so called because it is the origin point for the X, Y, and Z axes. At the origin point, the values of all the axes are zero.

When using the Move and Rotate tools, by default the tools will always work along the axes of the model space. By enabling Local Origin, the axes get aligned with the object, allowing it to be modified on the X, Y, and Z axes relative to the object, not the entire model space. Scale always uses the Local Origin.



MOVE WITH LOCAL ORIGIN

Click on {Move}

To enable

Find the fixture that is already rotated, and click on it.

Notice the X/Y/Z arrows are aligned with the origin axes of the model space

Click on {Local}

To enable the feature

Move the object.

Notice X/Y/Z are aligned with the rotated axes of the object, not the global axes

ROTATE WITH LOCAL ORIGIN

Click on {Rotate}

Notice that the axes align with the rotation of the object.

Rotate the object.

Notice that it travels along the rotated axes of the object, not the global axes.

Click on {Local} again

To toggle off the Local Origin feature

Rotate the object.

Notice that you can only rotate it along the model space axes.



Adding Library Objects

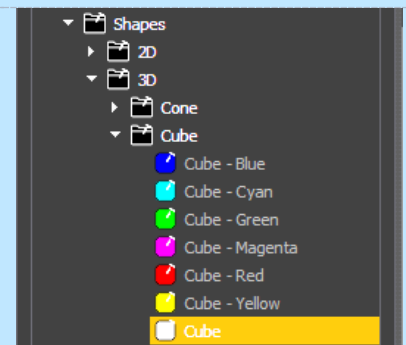
The Library is a collection of objects that can be brought into the model. You can add scenic and representational objects from the library into your Augment3d Model, giving you more surfaces with which to interact.

ADDING A 3D SHAPE

ADDING A CUBE

In Edit Mode, find the Library window on the far left

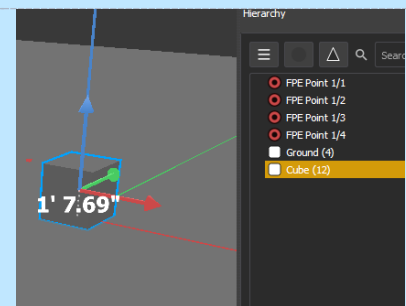
In the Library tree, go to Stock > Shapes > 3D > Cube > Cube (whichever color you'd like)



Click & Drag the Cube into the Model space.

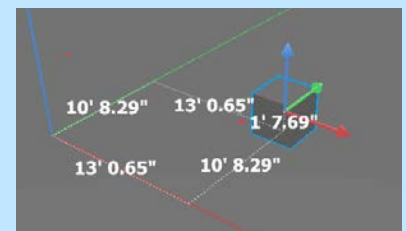
It appears in the model at XYZ: 0, 0, 0.

It also appears in the Hierarchy.

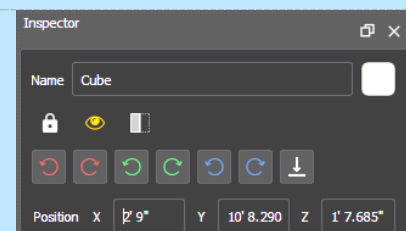


MOVE THE CUBE

With the Move tool selected, click & drag on the X, Y, or Z arrow handles to move the cube along that axis.



In the Inspector window, find the Position X field, and type [10] [Enter]

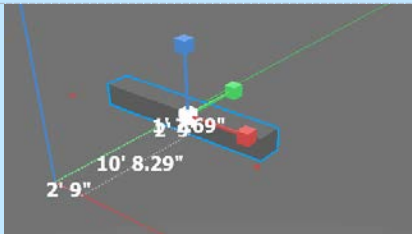


SCALE THE CUBE

With the Scale tool selected, click & drag the X, Y, or Z cube handles to scale the cube along that axis.

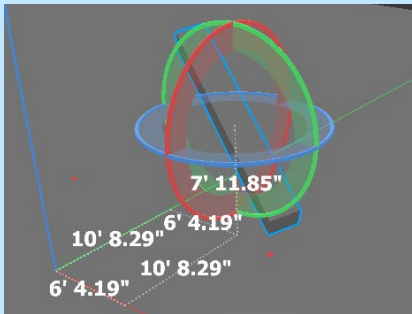


Click & drag on the white cube handle to proportionally scale the cube in all directions.

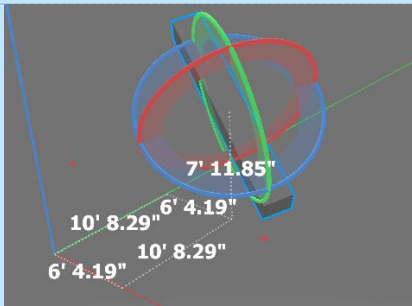


ROTATE THE CUBE

With the Rotate tool selected, click & drag the X, Y, or Z planet rings to rotate the cube along that axis.



- Click the Local Origin tool to enable.
- Notice the planet rings align with the rotation of the object.
- Rotate it in that orientation.



Click the Local Origin tool to disable it.

TO DELETE AN OBJECT

In the Library tree, go to Stock > Drape > Curtain Black

Click & drag the Curtain into the Model space, scale it up a bit.

To add something

In the Hierarchy, right click on the object in the menu

Choose Delete, then {Yes}

Removes object from the model

In the model, select the object and press [Delete]

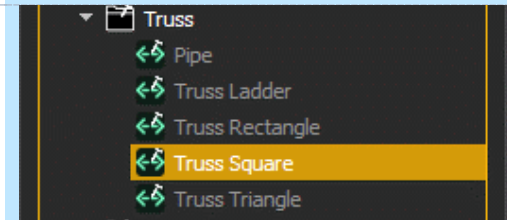
Note that the object is still in the Library, but is no longer in the Hierarchy

ADDING LIBRARY TRUSS

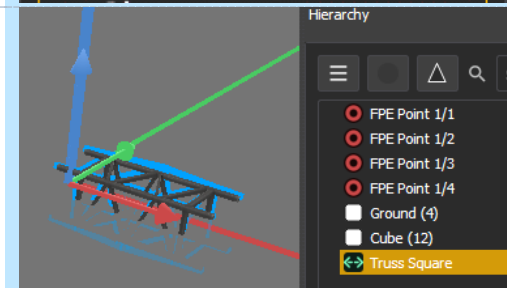
ADDING A TRUSS

Still in Edit Mode, find the Library window on the far left

In the Library tree, go to Stock > Truss > Truss Square



Click & Drag the Truss into the Model space. It appears in the model at XYZ: 0, 0, 0. It also appears in the Hierarchy.

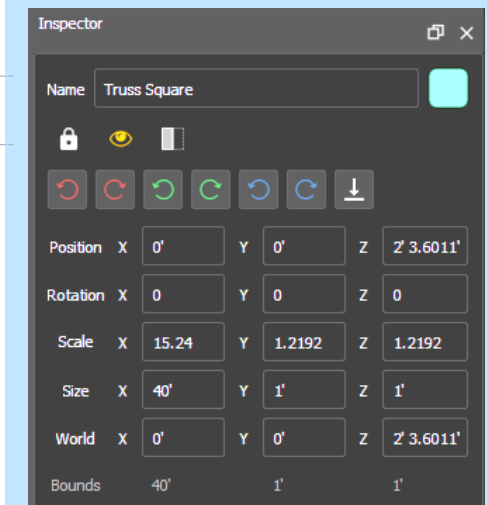


With the Move tool selected, drag the Truss up along the Z axis (blue arrow) to get a better view. Notice where the base point is located.

SIZING THE TRUSS

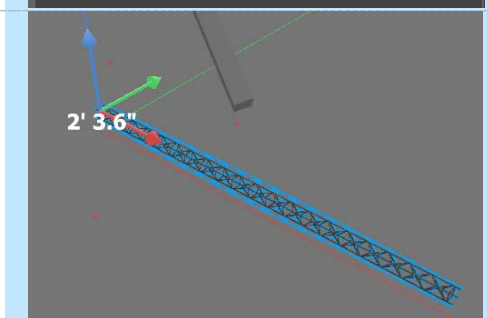
With the Truss still selected, look at Size in the Inspector window

Set the values to X = 40'-0", Y = 1'-0", Z = 1'-0"



Notice the Scale fields are connected to size – when we change one, the other updates.

Notice the truss grew in length away from the base point. We want it centered along the X axis.

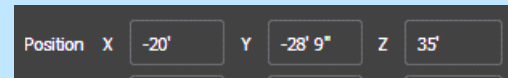


POSITIONING THE TRUSS

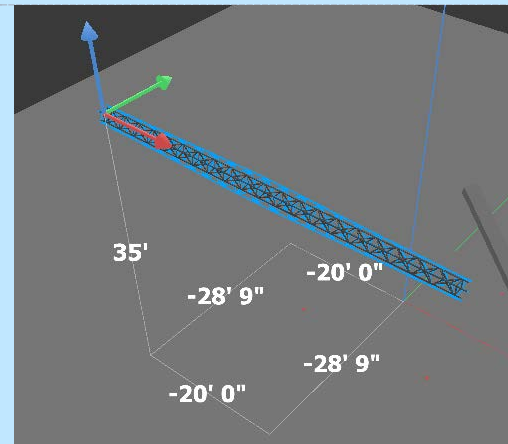
With the Truss still selected, look at Position in the Inspector window

Set the values to X = -20'-0", Y = -28'-9", Z = 35'-0".

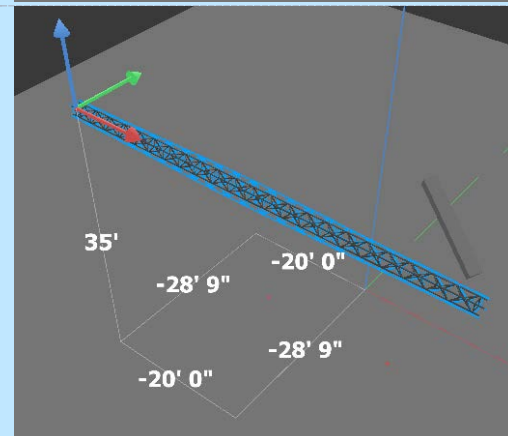
Be aware of the negative values for X and Y!



The truss is now centered on the X axis.



Change the Size X value to 45'-0". Notice the truss grows away from the base point, and is no longer centered.



Change the Size X value back to 40'-0"

BEST PRACTICE IDEAS

- The objects in the stock library are there to help you get representational objects in your model. They are a way to get objects in place so that light can bounce off of them and interact with them without necessarily being 100% accurate.
- Rather than drawing or importing highly detailed models for your venue and scenery, you can use simpler shapes to represent those objects, and interact with them. For example, do you need a fully detailed drum kit, or will a cube of similar volume get close enough?



Importing 3D Objects

Perhaps you've spent a lot of time drawing that perfect model of your space in a drafting program. Augment3d has tools that will allow you to utilize that work and not have to draw everything again.

Augment3d supports over 50 3D file formats. You can import individual objects, or an entire venue. Consider how you will want to interact with and move objects that you import, to decide whether you bring things into the model individually or as a group.

PREFERRED 3D OBJECT FILE FORMATS

Collada (.dae) and **Filmbox (.fbx)** – they have the most useful 3D information stored within the file, including orientation and origin.

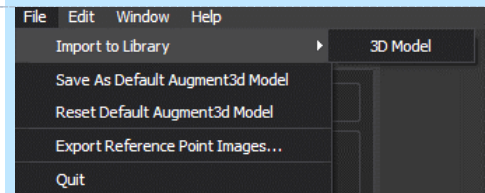
IMPORTING 3D OBJECTS INTO LIBRARY

ON A TETHERED COMPUTER

On a Tethered computer, you need to import 3D objects through the Tethered Edit Mode.

On the Tethered computer, insert a USB drive with the files you wish to import.

In Edit Mode, go to File > Import to Library > 3D Model



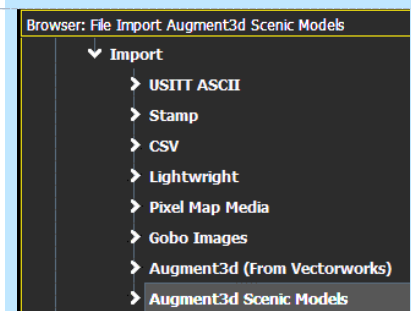
Browse to the file you want to import, and click Open. You will then see the Import Options Box.

ON A CONSOLE WITH AN AUGMENT3D TAB & ETCNOMAD

On a console with an Augment3d tab open, and in ETCnomad, you need to import 3D objects through the Browser.

On the console or in ETCnomad, insert a USB drive with the files you wish to import.

[Displays] > {File} > {Import} > {Augment3d Scenic Models}



Browse to the file you want to import, and double click or hit [Enter]. You will then see the Import Options Box.

SELECT THE FILE: "MIND THE GAP – SIMPLIFIED VENUE – REV A.DAE"

IMPORT OPTIONS BOX

The Import Options Box has settings for how the file gets imported into the Augment3d Library. In the graphic below, the Advanced button has been clicked, to show all available options.

Import Options for Mind The Gap - Simplified Venue - Rev A.dae
Scale (Feet/Inches)

Metres (122' x 91' 9.71" x 120') ▼

Original	W	122'	H	91' 9.71"	D	120'
Scaled	W	122'	H	91' 9.71"	D	120'

☒ Preserve Aspect Ratio

Origin Position

☐ As Model
 ☐ Center
 ☒ Ground
 ☐ Top

Model Orientation

☒ Automatic
 ☐ Z-Up

Optimization

☐ Combine Meshes
☒ Remove Small Meshes Min. Mesh Size: 1.18"

Advanced...

OK Cancel

SCALE

- **Scale Dropdown** – allows you to select the measurement used in the source file. If the source file has a defined measurement, this will be defined for you. Check the measurements to see if they look approximately correct.
- **Original Size** – shows the size in the original file. These fields are not editable.
- **Scaled Size** – allows you to scale the imported version of the file, and whether to preserve the aspect ratio between fields.

ORIGIN POSITIONS

- **As Model** – when the object is moved from the library into the model, the origin in the source file will be positioned at Augment3d's origin, which may result in objects being offset from the origin. This is a good choice if you are importing multiple objects from the same file (like Vectorworks) that need to align with each other after import.
- **Center** – when the object is moved from the library into the model, the object will appear with its X/Y/Z centered at the Augment3d origin.
- **Ground (default)** – when the object is moved from the library into the model, the object will appear with its X & Y centered on the Augment3d origin, but the lowest Z point of the object will be set at the Augment3d origin.
- **Top** – when the object is moved from the library into the model, the object will appear with its X & Y centered on the Augment3d origin, but the highest Z point of the object will be set at the Augment3d origin.

MODEL ORIENTATION

- **Automatic** – will use the orientation information in the source file to determine the orientation after import
- **Z-Up** – will force the Z orientation to match the Augment3d Z axis.

OPTIMIZATION

- **Combine Meshes** – Upon import, Augment3d will try to simplify the objects triangles by combining similar meshes.
- **Remove Small Meshes** – upon import, Augment3d can remove meshes below a certain size, which you can define.

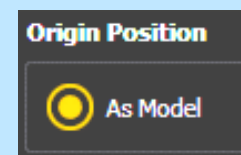
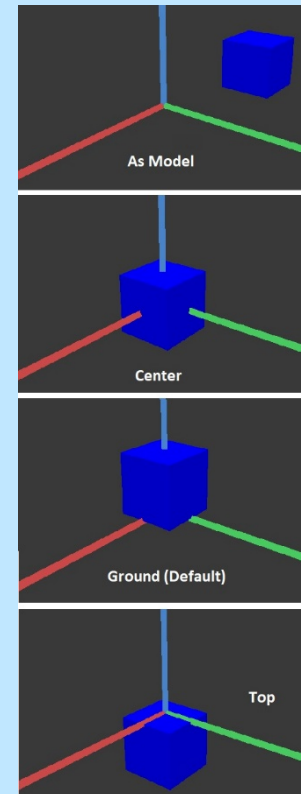
For importing "Mind The Gap – Simplified Venue – Rev A.dae,"

Leave the scale at meters with no scale changes,

Leave Orientation and Optimization at defaults.

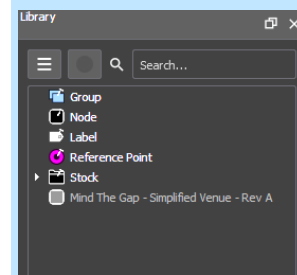
Change Origin Position to "As Model"

When the settings are correct, click OK

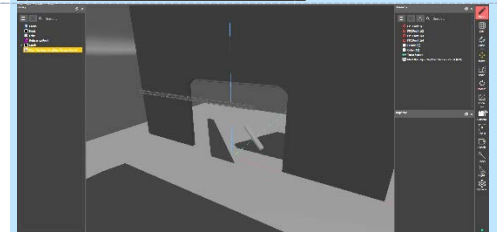


MOVING FROM LIBRARY TO MODEL

Once you have completed an import, the object is available for use in the library.



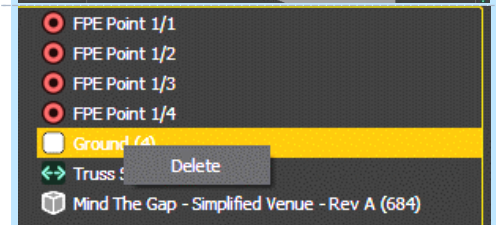
To place the Venue into the model, click & drag from the library into the model space. It doesn't matter where you drop it, the object will use the origin set from your import preferences.



In the Hierarchy, right-click on the Cube, and delete it.

Right-click on the Ground, and delete it.

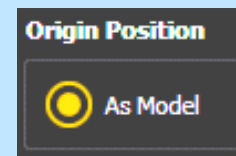
We can leave the Truss, we will use it, and it is in the proper position.



IMPORT 3D SCENERY & MASKING

Navigate to import "Mind The Gap – Scenery & Masking – Rev A.dae"

In the Import Options Box, make sure Origin Position is set to "As Model."

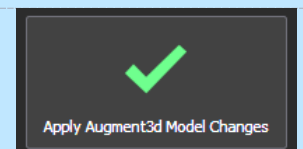


Click OK.

Drag & Drop the object from the library into the model.

Notice that the scenery lines up with the venue, because they both came from the same Vectorworks file, and we used the source file's origins for import.

Click on {Done}, and {Apply Augment3d Model Changes}





Saving & File Management

SAVING & EOS SHOW FILE TYPES

There are several options for saving a show file.

Make sure you are out of Augment3d Edit Mode, and hit [Live]

[Displays] – Browser, select File > Save As > Show File Archive

- **Save with Augment3d Model (.esf3d) (Default)** – saves all show data, as well as the model objects in the file. File sizes will be larger depending on the complexity of the Augment3d model.
- **Save Show Data Only (.esf2)** – saves only the show data and does not save the model objects. The Fixtures' Augment3d information, stored in patch, will be saved with this file type.
- **Save in Legacy Mode for Eos v2.9.x and Lower (.esf)** – if you created a show file in 3.0 and need to open it in an earlier version of Eos software, you can save in Legacy mode.

Keep {Save with Augment3d Model}, selected and hit {OK}.

You will be prompted to name the file – do so, and then hit [Enter]

QUICK-SAVE AND FILE TYPE

Executing a Quick-save by using **[Shift] & [Update]** will save in the last file format you selected, between .esf2 and .esf3d.

For example, if you did a Save As and selected .esf2, your quick-saves will continue to be in .esf2 format until you Save As and select a different file type.

The exception is legacy file Save As (.esf) – the save in this file type is always one-off, and the quick-save after will continue to use the previously used file format.

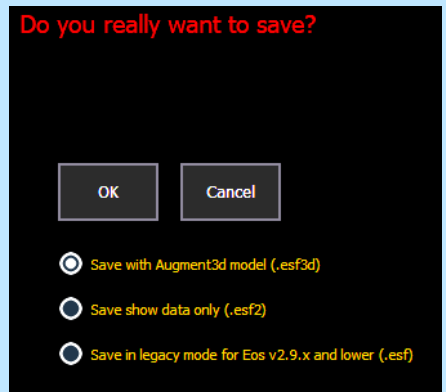
UNSAVED CHANGES INDICATORS

There are different indicators to show whether unsaved changes have been made to the Show Data, the Augment3d model, or both.

- Unsaved changes only in the Show Data.
- Unsaved changes only in the Augment3d model.
- Unsaved changes both in the Show Data and the Augment3d model.
- No unsaved changes

BEST PRACTICE IDEAS

- Because Augment3d model data can take up a lot of space, if the model isn't changing while cueing or editing other show data, you can save in .esf2 format throughout the day, and then save a full .esf3d file at the end of the day in order to save space.



Eos Family Expert Topics - File Types (Augment3d*)
Eos Family Expert Topics - File Types* (Augment3d*)
Eos Family Expert Topics - File Types
Eos Family Expert Topics - File Types*

BEFORE CONTINUING, OPEN THE SHOW FILE:

"EOS FAMILY LEARNING SERIES – EXPERT TOPICS – REV A.ESF3D"



Click to Select

In Augment3d, we can select channels simply by clicking on the fixtures in the model. These selections are fully interactive with the command line.

In the Model Space, Camera menu, click {Home}.

Adjust your camera back so you can see the FOH Truss.

Click on the right-most fixture, Channel 101.

The channel appears on the command line. The fixture highlights in a gold border, just like a channel tile in the Live Channel display, indicating the channel is selected.

Click the next mover to the left (skipping the Source 4s), Channel 102, then 103, then 104.

Channel 102 is added to the command line selection. The fixture also highlights gold.

On the keypad, [+] [105] [Enter]

You can use the command line interchangeably with Click selection, just like in a Live Channel display.

This is advantageous in Augment3d, as some fixtures may not always be visible to click select.

Click on a fixture (104)

Removes that channel (104) from the selection

Double click on any fixture in the Model Space

All other channels are deselected. Only the selected fixture remain

Double click on any non-fixture area

Deselects all channels.

Click and drag a box around the truss to select all the fixtures

Notice the command line populates the channel selection.

[Full] [Full]

All the fixtures turn on

Depending on your view – you may have to move your camera angle to not include other fixtures in your selection box.

Double click on any non-fixture area

Deselects all channels.

AUGMENT3D SELECTION COLOR SIGNIFIERS

Depending on how a channel is selected, it highlights with different colors.

- If a channel is selected on the Command Line, it is highlighted in gold.
- In Augment3d Edit Mode, if a channel is selected with the Move, Scale, or Rotate tool, it is highlighted in blue.
- In Augment3d Edit Mode, if a channel is selected on the Command Line and the Augment3d tools, it is highlighted in magenta.





Click to Focus

Click to Focus makes moving one or several moving lights to a point on stage easy, and much faster than driving each unit by its encoders.

AUGMENT3D CLICK TO FOCUS

[Clear] [Sneak] [0] [Enter]

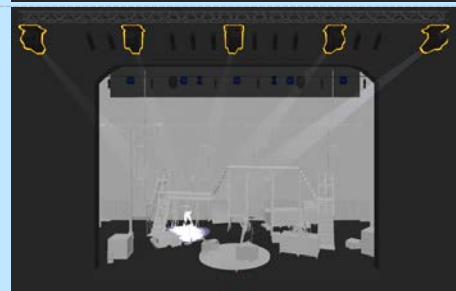
Clear all the manual values.

Click to Select the movers on the FOH truss or type [101] [Thru] [105] then [Full] [Full]

Click on the top of the crate under the downstage singer's feet & hold for 1 second



Click on the top of the car under the guitar player & hold for 1 second



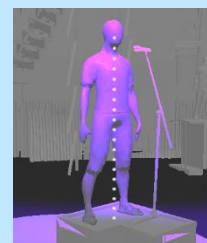
Z OFFSET – MOUSE WHEEL

Click to Focus speeds up moving multiple movers to a single point on stage, but we are not often lighting people's feet. You can set a Z-Offset, so that when you click to focus on the top of a surface, the fixtures point above that location by a defined measurement.

Click and hold the left mouse button and roll the mouse wheel up.

Depending on your camera view, you may see a dotted line moving in the Z axis above your click point. The focus of the lights will move up with the top of the dotted line, allowing you to focus higher than your Click to Focus point.

Once the lights are focused on the guitar player's face, let go of the left mouse button.



This sets the Z offset value

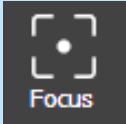
Now Click to Focus on the crate under the singer

That value is maintained as you continue

Focus Offset is maintained until reset by rolling down the mouse wheel, resetting the Focus Offset value to zero (described below), or by playing a snapshot that has a different Focus Offset stored in it.

FOCUS MENU

The Focus Menu contains Focus Offset and Movement tools.

Click on {Focus} Menu	
In Focus Height Offset, click and type [5] [.] [6] [Enter]	
Click to Focus on the drummer's crate and on the car/guitar player	
Click on {Focus} Menu, {Animate Pan/Tilt} to disable the animation	
Click to Focus on the drummer again	Beams snap into place, rather than move
Click on {Focus} Menu, {Animate Pan/Tilt}	To enable the animation again
Click to Focus on the singer again	Beams move into position

BEST PRACTICE IDEAS

Having more objects in your model gives you more interaction points, which provides better accuracy and saves time when using Click to Focus.

Advanced Displays & Control

In this section, we will be looking at Augment3d displays and control menus, including Labels, Sticks, Light, and Options.

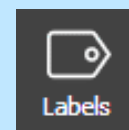


LABELS MENU

Labels are used to display channel numbers at the fixture, mid-beam, or at the end of the beam. You can use multiple types at the same time, and there are multiple modes that determine when a label is shown for a channel.

They can also be clicked on for channel selection, just like click-to-select for fixtures. Their text color indicates the state of the channel (such as manual, stored, submaster, etc.).

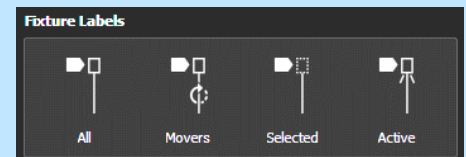
On the Augment3d toolbar, click on the {Labels} menu.



FIXTURE LABELS

Fixture labels will show the channel number next to the fixture location. Fixture labels will be visible through objects, so you can see channels behind architecture and scenery.

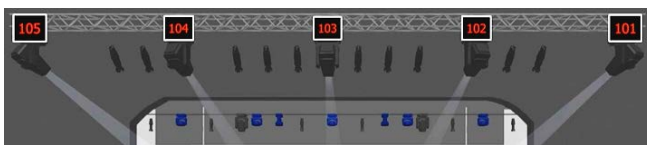
- **All** – Fixture labels will appear on all channels shown in Augment3d.
- **Movers** – Fixture labels will appear on only moving lights.
- **Selected** – Fixture labels will appear on channels that are selected.
- **Active** – Fixture labels will appear on channels that are active.



ACTIVE

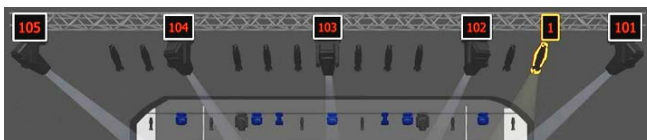
Make sure 101 through 105 are still at full and that you can see them

Click on {Labels}, under Fixture Labels, {Active}



You can now see the fixture labels for the just active channels, 101-105.

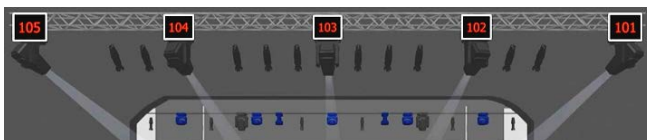
[1] [At] [Full] [Enter]



Because channel 1 is now active, the channel label appears.

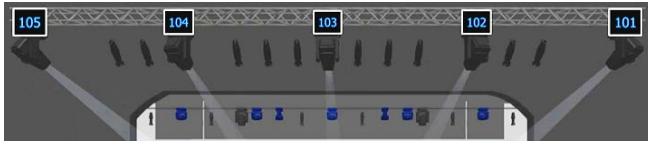
Remember: Red is manual data

[1] [Sneak] [Enter] [Clear]



Channel 1's label now disappears because it is no longer active.

[Record] [201] [Enter]



The channel labels turn to blue, because the channel data is now stored in a cue, just like in Live Channel displays.

The label will also be in yellow if data is coming from a submaster and orange if coming from a playback fader.

Keep in mind we are showing you the state of the intensity parameter of these fixtures. If you have different data in different states on other parameters, those won't be displayed.

SELECTED

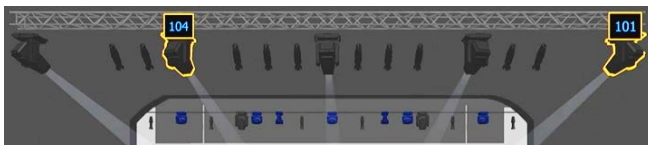
Click on {Labels}, under Fixture Labels, {Active}.

A toggle to enable -disable

Click under Fixture Labels, {Selected}

To turn on labels for Selected channels.

[101] [+] [104] [Enter]



Channels 101 and 104's labels are displayed, because they are on the command line.

[Clear]

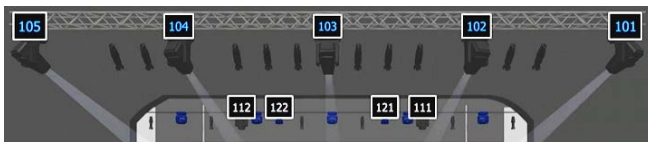
The labels go away, as no channels are selected.

MOVERS

Click on {Labels}, under Fixture Labels, {Selected}.

To turn off Selected

Click under Fixture Labels, {Movers}



All movers show their channel numbers, and label colors are indicative of the state of the channel.

Click on {Labels}, under Fixture Labels, {All}.

Shows all channel labels. Notice how cluttered the view can get.

Click on {All} again

Turns off all channel labels.

BEAM CENTER LABELS

Beam Center Labels put the channel number at the center of the beam. Because they can be used for channel selection, these are great for selecting the channel when you can't see the fixture.

- **All** – Beam Center labels will appear on all channels shown in Augment3d.
- **Movers** – Labels will appear on only moving lights.
- **Selected** – Labels will appear on channels that are selected.
- **Active** – Labels will appear on channels that are active.

Click on {Labels}, under Beam Center Labels, {Movers}.

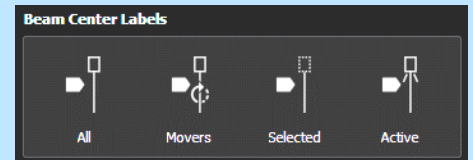


If fixtures are hidden and can't click to select them:

Click on the label for channel 111.

Click on {Labels} under Beam Center, {Movers}.

[Shift] & [Clear]



All moving lights display their channel number in the middle of the beam.

Channel 111 is selected on the command line.

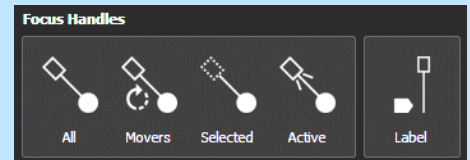
Turns off those labels

Clear the command line.

FOCUS HANDLES & LABELS

Focus Handles are a special kind of beam label, in that they allow you to reposition the light via click & drag. They appear at the center of a beam where it intersects with objects, and follow object contours. They are also useful for creating and maintaining focus arrangements when they need to move.

- **All** – Focus Handles will appear on all channels shown in Augment3d.
- **Movers** – Focus Handles will appear on only moving lights.
- **Selected** – Focus Handles will appear on channels that are selected.
- **Active** – Focus Handles will appear on channels that are active.
- **Label** – Fixture labels appear near the focus handle.

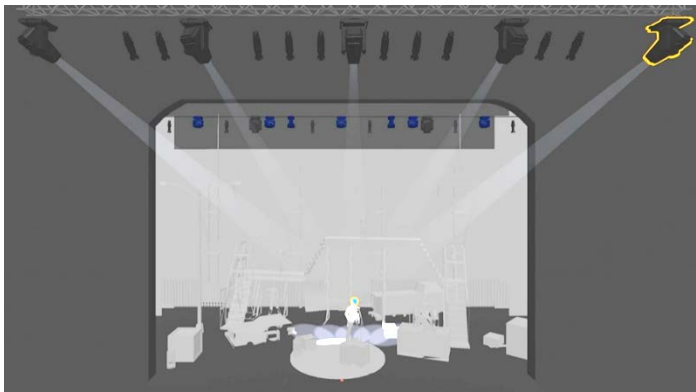


Click on {Labels}, under Focus Handles, {Selected}

to turn on focus handle labels.

You should see a white dot appear on the floor.

[Clear], [101] [Enter]



A blue focus handle appears where the center of the light is focused, currently on the singer's head.

Click & drag the beam to downstage left.

Notice the focus handle turns red,

Console understands where the light is intersecting with the model. The dot turned red because the fixture information is now manual.

[101] [Thru] [105] [Home] [Enter]

Homes the FOH movers.

[Next], click and drag 101 downstage left

[Next], click and drag 102 middle left

[Next], click and drag 103 upstage center

[Next], click and drag 104 middle right

[Next], click and drag 105 downstage right



Get all of the channels into a "V"

[Select Last] [Enter]

Selects 101-105

Click & drag any of the focus handles

The channels maintain their arrangement relative to each other.

Click on {Labels}, under Focus Handles, {Labels}



Shows which dot belongs to which channel

Click on {Labels} {Restore Label Defaults}

Resets the labels, turning them all off.

BEST PRACTICE IDEAS

- Labels can be used for channel selection – a great way to interact with a fixture you can't see because it is behind scenery or out of your camera view.



STICKS MENU

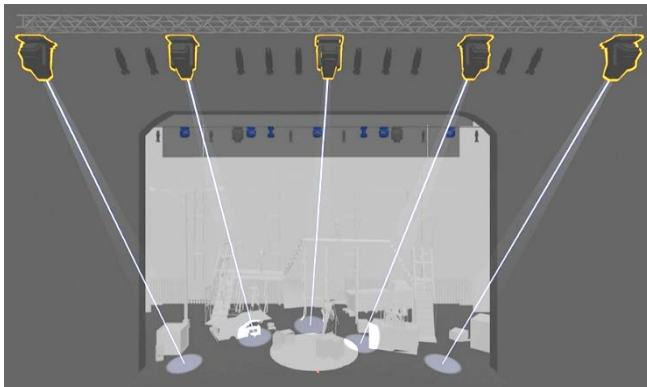
Stick Beams draw a simple line down the center of the beam from the fixture to where it intersects with an object. Like Labels, you can use them on All Channels, or only on Moving Lights, Selected Channels, Active Channels, or Inactive Channels. The Stick Beam will display the color and intensity of the channel.

On the Augment3d toolbar, click on the {Sticks} menu.

- **All** – stick beams will appear on all channels shown in Augment3d.
- **Movers** – stick beams will appear on only moving lights.
- **Selected** – stick beams will appear on channels that are selected.
- **Active** – stick beams will appear on channels that are active.
- **Inactive** – stick beams will appear on channels that are not active.

[101] [Thru] [105] [Enter]

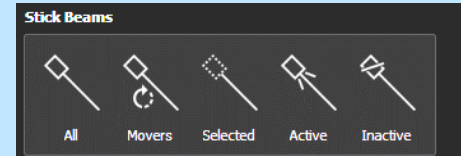
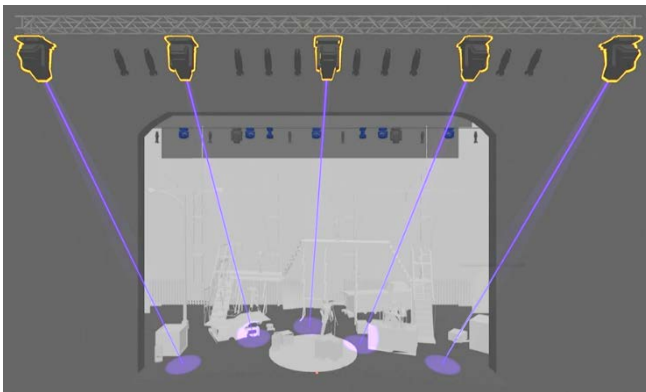
Click on {Sticks}, click on {Selected}



[Clear]

[Select Last] [Enter]

On the Color Encoders, put {Cyan} and {Magenta} to Full



The stick beams will run through the center of the beam, and will show the channels' intensity and color.

Clear the selection, stick beams go away

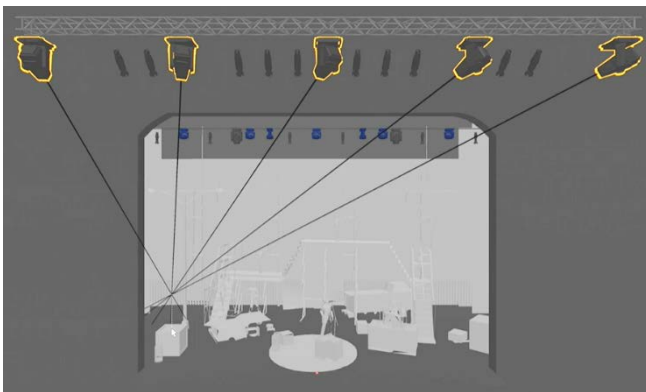
Grabs Channel 101 thru 105 again

The stick beams change to the color of the channel.

[Select Last] [Out] or role the level wheel down

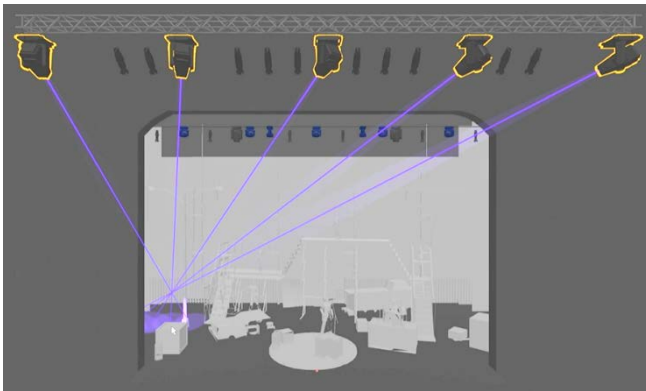
Takes out the intensity, still have the sticks

Click & hold on the top of the crate downstage right.



Notice the sticks stay in the beam, but are now black. You can Click to Focus the channels to a new location.

[Select Last] [Full] [Full] or roll the level wheel back to full



The channels are now lit up in their new location.

Click on {Sticks} then on {Selected}

Toggles off the stick beams.

BEST PRACTICE IDEAS

- For a low-resource Augment3d session, enable stick beams, and in Patch > Augment3d, have {Hide Beam} enabled.
- Stick Beams on inactive channels can show where fixtures are located when they are not being used, for better understanding of fixture tracking and marking.



LIGHT MENU

The Light menu gives you options to change the quality of the light atmosphere in the Augment3d model.

On the Augment3d toolbar, click on the {Light} menu.

BEAMS

These settings allow you to choose which beams get rendered through the air. All beams are on by default. With them off, you still see the beams on the objects, but the cone of light through the air is not rendered. This can save on processing power on your console or computer.

- **All** – stick beams will appear on all channels shown in Augment3d.
- **Movers** – stick beams will appear on only moving lights.
- **Selected** – stick beams will appear on channels that are selected.
- **Intensity** – allows you to offset the brightness of the beam where it falls onto objects. The default is 50.

Under Beams, click on {All}

Click on {Light}, under Beams, click {All} again

LIGHT & BEAM INTENSITY

These allow you to adjust the light so that it is more representable of the light in your space.

- **Beam Intensity** – adjusts the brightness of the cone of light through the air. (Default is Medium)
- **Light** – adjusts the brightness of the light (Default is 50%)

Click through the options to watch the beams change.

HAZE

Haze allows you to introduce volume to the air which makes the beams more visible. A simple toggle on or off.

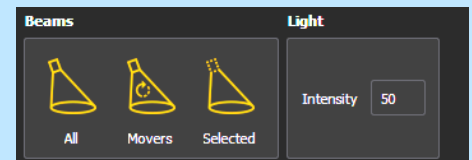
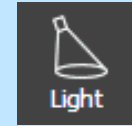
AMBIENT LIGHT

These settings allow you to control the ambient light in the model. It is what lights the objects that are not being lit by luminaires

- **Off** – only see the things that are illuminated by fixtures.
- **Low, Medium, High** – you can adjust how bright the ambient light is for the model (medium is the default).

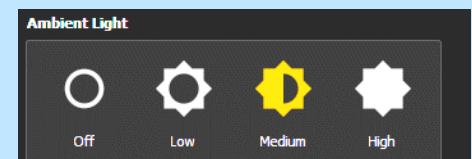
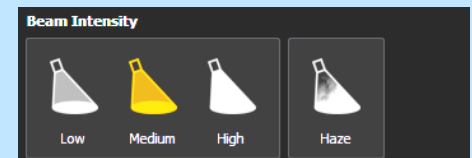
BEST PRACTICE IDEAS

- Many of these settings can help you quickly improve graphics performance. Things like turning off beams and haze make processing the model easier.
- All these settings are stored when you record a Snapshot. There is currently no way to break these settings out individually.



Turns off all beam cones. But you can still see the beams on the scenery and floor.

To turn the beams back on.





OPTIONS MENU

The Options Menu gives access to Cue Preview Mode, and the Augment3d graphical quality settings.

On the Augment3d toolbar, click on the {Options} menu.



CUE PREVIEW MODE

The Augment3d model will display either the current cue or, when Cue Preview Mode is enabled, the pending cue from the Master Playback.

Click off the Options window, [Clear]

To clear command line

[Go To Cue] [101] [Enter]

Puts Cue 101 on stage, the model shows the output of Cue 101 on stage

Click on {Options} {Pending Cue}

Enables Pending Cue. The model is now what is coming up in Cue 102.

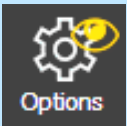
[▶] (Go) . [▶] (Go) ...

You will continue to see the pending cue in the Augment3d window.

Click on {Options} {Pending Cue} to disable

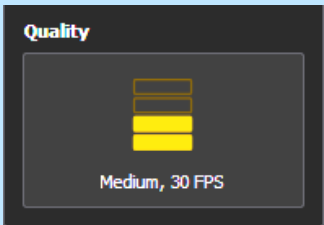
The Augment3d scene returns to showing the current cue.

When Cue Pending is enabled, the Options menu icon shows the Cue Pending eye overlaid.



QUALITY

The other setting in the Options menu is Quality, which includes graphical settings for the Augment3d model space. Quality is covered in detail in an earlier chapter in this book, titled “Display Quality Settings.”





Augment3d & Snapshots

For quick recall of Augment3d camera positioning, beam attributes, and atmospheric settings, use Snapshots.

Using your mouse or keypad tools, move the camera to an upper-right view of the stage.

[Record] [Snapshot] [401] [Label] Balcony Right [Enter]

Snapshots record camera pose settings.

Now move the camera to an upper-left view of the stage.

[Record] [Snapshot] [402] [Label] Balcony Left [Enter]

Toggle between Snapshot 401 & 402

The camera moves when the snapshot is played.

We also save all display settings in Snapshots.

Turn on Stick Beams for just the Movers

Adjust camera to a slightly different position

[Record] [Snapshot] [403] [Label] Sticks [Enter]

Snapshots include all camera pose properties, as well as atmosphere and beam settings.

Toggle between Snapshot 401 & 403

The camera moves, and beam settings are updated.

BEST PRACTICE IDEAS

- Snapshots are just like they are anywhere else in the console, an easy way to quickly store your display settings and recall them quickly. This includes display tools, such as Sticks and Labels, beam brightness, haze, and ambient light settings.
- Currently, the Display Quality Settings are recorded into snapshots in addition to camera and atmosphere settings. This can cause difficulty when moving a file to a new system and recalling snapshots on machines with differing processing capabilities. For this reason, it may be good to record snapshots that only affect the Augment3d tab.



Focus Palettes – XYZ Mode

Focus palettes can now contain XYZ data. When using this type of focus palette, the console stores a position in space rather than only the pan and tilt data. This allows the light to consistently hit the same point regardless of the physical location of that fixture.

USING X, Y, & Z PARAMETERS FOR FOCUS

When controlling a group of movers, you can move the location that they are pointing at in space, as opposed to just pan & tilt.

[Clear] [Sneak] [Enter]

In the Augment3d window, click on {Camera}, {Bookmark 1}.

Click on {Sticks} and turn off all Stick Beams.

[101] [Thru] [105] [At] [Full] [Enter]

Click To Focus the fixtures on the feet of the guitar player

With Z value, lights are on the head of the guitar player

[Record] [Focus Palette] [401] [Enter]

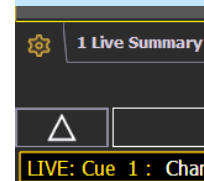
Records the channels in a focus palette.

[Select Last], move pan/tilt encoders

Fixtures move as they always have.

Access the Encoder Display:

- On a Ti: look at the Pan/Tilt encoder display area
- On other consoles: [Encoder Display] to open the Encoder Display in the CIA
- On ETCnomad: click the Encoder Display Arrow to open the Encoder Display in the CIA



Access the X, Y, & Z Encoders:

- On Ti: Touch {Switch to XYZ}
- On other consoles & ETCnomad: change Focus encoder pages using the [{Focus}] button

The X, Y, and Z encoders will be visible

Roll the {X} encoder to move all fixtures in a straight line across the stage

Look at the Pan/Tilt values and the X/Y/Z values in the Live Table display

Roll the {Y} encoder to move all fixtures in a straight line up and down the stage

Using Pan/Tilt to make the compound moves to maintain the line

Using the {X} & {Y} encoders, move the lights to the bottom of the center box (where the singer is located)

Using the {Z} encoder, move the lights up to focus on the singer's head

Note, in Live display, that the pan and tilt values are red (manual) and, even though XYZ values are changing, they remain gray.

CHOOSING PAN/TILT OR XYZ FOCUS SPACE

You can store Cue and Focus Palette data in Pan/Tilt focus space or XYZ focus space. When stored in Pan/Tilt space, the fixture will apply the stored values to those parameters, regardless of where the fixture is located in space. When stored in XYZ focus space, the fixture will point at the stored location in space, regardless of where the fixture is located.

In a Live or Blind display, the used space will be manual/stored, while the unused space will be grayed out, but they are always tied together.

Pan	Tilt	X Focus	Y Focus	Z Focus
-37	-49	-5' 11"	5' 7"	-3' 6"
-22	-45	-3' 8"	6' 2"	-3' 6"
-2	-43	-1' 3"	6' 7"	-3' 6"
18	-45	1' 10"	6' 11"	-3' 6"
35	-49	4' 11"	6' 11"	-3' 6"

CONVERT TO XYZ SPACE IN LIVE

[Select Last] [Focus Palette] [401] [Enter]

[Blind] [Focus Palette] [401] [Enter]

Notice the Pan/Tilt values are white – that is what is stored in the palette, and will be applied when it is used.

[Live] [Select Last], click to focus on the drummer

On the Focus encoder page, {XYZ Format Enable} will change from Pan/Tilt focus space to XYZ focus space

This does not move the lights at all but it is going to change the red data from Pan/Tilt to XYZ indicating that that is how the data will be stored now.

[Record] [Focus Palette] [402] [Enter]

Now whenever Focus Palette 402 is used, if the fixtures move in real space those fixtures will still be focused on the same area on our stage. The light will point at the drummer, regardless of where the light is located.

Places channels 101 thru 105 back into Focus Palette 401

Ch	Pan	Tilt	X Focus	Y Focus	Z Focus
101	-34	66	-11' 4"	17' 7"	8' 1"
102	-25	63	-11' 4"	17' 7"	8' 1"
103	-14	62	-11' 4"	17' 7"	8' 1"
104	-2	61	-11' 4"	17' 7"	8' 1"
105	10	61	-11' 4"	17' 7"	8' 1"

Pan	Tilt	X Focus	Y Focus	Z Focus
-13	56	11' 4"	7' 5"	9' 1"
3	56	11' 4"	7' 5"	9' 1"
18	57	11' 4"	7' 5"	9' 1"
31	60	11' 4"	7' 5"	9' 1"
41	63	11' 4"	7' 5"	9' 1"

Pan	Tilt	X Focus	Y Focus	Z Focus
FP 402	FP 402	FP 402	FP 402	FP 402
FP 402	FP 402	FP 402	FP 402	FP 402
FP 402	FP 402	FP 402	FP 402	FP 402
FP 402	FP 402	FP 402	FP 402	FP 402
FP 402	FP 402	FP 402	FP 402	FP 402

Let's move the fixtures' position to see how this works.

[Patch], [101] [Thru] [105] [Enter]

Go into Patch, and select the fixtures

{Augment3d}

Go into the Augment3d section

{Position} [/] [/] [+] [-] [8] [Enter]

[/] [/] skips the X and Y fields, allowing you to give only a Z value. [+] [-] allows for a negative offset from the stored value

[Live]

The fixtures are still pointed at the drummer, even though they are in a different location

When the position was changed in Patch, the console auto-calculated the Pan/Tilt values to compensate for the change in the XYZ coordinates of the fixtures, keeping them pointed at the space place on stage.

CONVERT TO XYZ SPACE IN BLIND

[Blind] [Format], until you are in Spreadsheet

[Focus Palette] [101] [Thru] [109] [Enter]

Selects those nine Focus Palettes

[101] [Thru] [105] [Enter]

Selects the FOH mover channels

Notice the channels have values stored in Pan/Tilt focus space

		101				
Palette	Label	Pan	Tilt	X Focus	Y Focus	Z Focus
101	USR	-41	62	-12 2"	8 8"	0'
102	USC	-18	60	5 5"	14"	0'
103	USL	2	54	19 11"	6 9"	0'
104	CSR	-44	60	-11 1"	4"	0'
105	CS	-25	52	5 9"	1 3"	0'
106	CSL	2	50	19 10"	2 2"	0'
107	DSR	-50	56	-8 10"	-3 9"	0'
108	DSC	-30	47	5 6"	-3 8"	0'
109	DSL	-4	43	17 3"	-4 4"	0'

In the Encoder Display, {XYZ Format Enable}

All the channels are now stored to XYZ focus space

		101				
Palette	Label	Pan	Tilt	X Focus	Y Focus	Z Focus
101	USR	-41	62	-12 2"	8 8"	0'
102	USC	-18	60	5 5"	14"	0'
103	USL	2	54	19 11"	6 9"	0'
104	CSR	-44	60	-11 1"	4"	0'
105	CS	-25	52	5 9"	1 3"	0'
106	CSL	2	50	19 10"	2 2"	0'
107	DSR	-50	56	-8 10"	-3 9"	0'
108	DSC	-30	47	5 6"	-3 8"	0'
109	DSL	-4	43	17 3"	-4 4"	0'

We should return the FOH movers to their original position.

[Patch], [101] [Thru] [105] [Enter] {Augment3d} tab

{Position} [/] [/] [+] [8] [Enter]

Moves the channels up 8 feet on the Z axis, restoring them to their previous location.

[Live]

BEST PRACTICE IDEAS

- Pan/Tilt focus space is great for looks where you care about the beam location, not where it is pointed – like a group of deck fixtures making a fan of beams through the air.
- XYZ focus space is great for keeping fixtures pointed at the same location, even if the fixtures will be moving regularly – like Front of House movers on a tour. XYZ focus space is also great for live fading movers in a straight line.
- Fixtures will always use the destination data – so if a channel is fading from a cue stored in Pan/Tilt space into a cue stored in XYZ space, the fade will respect the XYZ data.

Staging Mode

Staging Mode lets you prepare a look in the Augment3d display, and then choose to commit it to Live. We can also use this powerful feature in Blind.

[Stage] is next to **[Blind]**. In 3.0, **[Scroll Lock]** was replaced by **[Stage]** – **[Shift]&[Stage]** will enable or disable scroll lock.



STAGING IN LIVE

[Live], [Go To Cue] [Out] [Enter]

In the Augment3d window, click on {Options}, {Quality}, and set the Ambient Light setting to {Low}

Adjust your ambient light to better reflect the real space

[Stage]

To go into Staging Mode

Staging mode appears on the leading edge of the Command line like other modes on the console. The LED on the [Stage] button is lit.



[101] [Thru] [105] [Full] [Enter]

Ch	Intens
101	FL
102	FL
103	FL
104	FL
105	FL

The channels turn on in the Augment3d model, but not on stage.

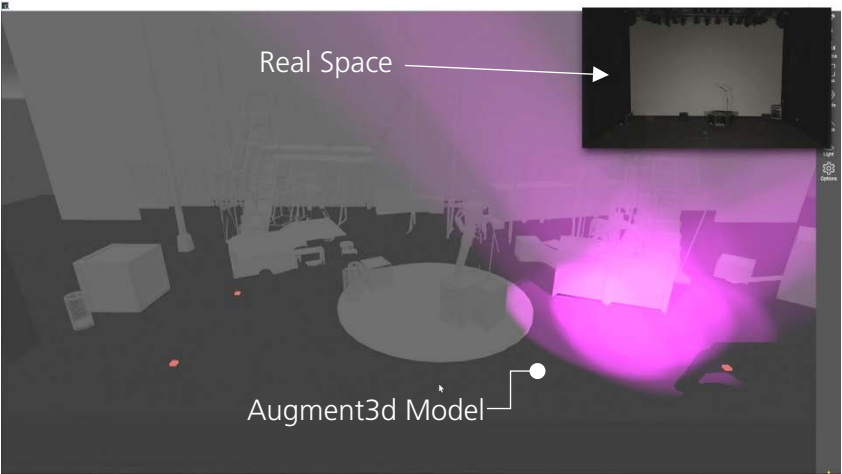
The Staging data is displayed in Orange.

In the Augment3d model, click to focus downstage of the drummer

Nothing happened in Live

Open a color picker and make them magenta

Still nothing in Live



TO CLEAR STAGING MODE

[Stage] [Stage] or [Stage] [Enter]

To clear staging mode and release the changes – Augment3d matches reality, and the orange data is cleared

[Go To Cue] [102] [Enter]

Go to cue 102

[Stage]

Enter into Staging Mode

[102] [+] [104] [Enter] – Click to Focus on Singer, zoom out a bit, make amber

Z height is maintained from before. This makes the changes in the model only



[102] [Copy To] [Enter]

Channel 102's data moves from the Staging buffer into Live Manual data. Changes snap in a zero time. Before [Copy To], the data was orange. After it is red, indicating manual in Live.



[Clear]

To clear the command line

[Copy To] [Sneak] [3] [Enter]

In Live display, note channel 104 is orange until press Enter

Using Sneak, you can bring changes from Staging Mode into reality over time. Because we used Copy To with no channel selection, all values in Staging Mode will be copied into Live.

[Stage] [Stage]

Clears Staging Mode

[Update] [Enter]

Update Cue 102



STAGING IN BLIND

[Blind]	
[Next] [Next] [Next]	Model previews the cues, while nothing is live on stage
[Cue] [30] [Enter]	
[Stage]	To go into Staging mode
[51] [Thru] [53] [+] [71] [Thru] [73] [Out]	Intensity changes show up in orange. They are in the Staging buffer, and have not been committed to Cue 30.
[Stage] [Stage]	Clears staging mode without keeping the changes
[Last] [Last]	Gets you into Cue 28
[Stage]	To go into Staging mode
[51] [Thru] [53] [Enter], and make them amber, [At] [50] [Enter]	Color and intensity values are orange in the Blind table view.
[Clear]	Clears the command line so there is no selection and can commit all changes
[Copy To] [Enter]	Commits all Staging changes to Cue 28. The values now show in stored colors.
[Stage] [Stage]	To clear Staging mode

Remember: Changes made in Blind are stored automatically.



USING STAGING BETWEEN LIVE & BLIND

Another great feature of Staging mode is that it allows you to hold data in the Staging buffer even if you are bouncing between Live and Blind.

BLIND > LIVE > BLIND

Still in Blind, [Cue] [13] [Enter]	
[Stage]	To go into Staging mode
[31] [Thru] [35] [Out], [71] [Thru] [73] {Color} [Recall from] [51] [Enter]	The designer hasn't finished making changes, but needs to go into Live to make some edits.
[Live]	You are in Cue 102 in Live. The Stage LED begins blinking, indicating there is something in the Staging Buffer. It will continue to blink until it is committed or cleared.
Press [▶] (Go) into cue 102.5	
[111] [+] [112] [Full] [Full]	Making the changes in Live
In the Augment3d window, Click to focus downstage of the singer, drop in a gobo and sharpen	
[Update] [Enter] [Time] [3] [Enter]	The manual changes made are updated to cue 102.5

[Blind], [Cue] [13] [Enter]	Notice the intensity of channels 31 thru 35 is Full.
[Stage]	Notice that Channels 31 thru 35 are now displayed in orange, and the data is being contributed from the Staging values.
[Next], [Next], [Last], [Last] , Back to cue 13	Info in Staging buffer is held and can be applied to any cue or target.
[51] [Thru] [53] [Out]	Adds these values to the Staging buffer
[Clear]	To clear the command line
[Copy To] [Enter]	Orange data changes to cue level data
[Stage] [Stage]	To clear Staging mode
 LIVE > BLIND > LIVE	
[Live], in cue 102.5	
[Stage]	To go into Staging mode
[102] [+] [104] [At] [75] [Enter]	
Click to focus on the Singer, drop in a gobo, zoom	Designer needs to edit something in Blind now
[Blind]	
[Next], [Next], [Next] to Cue 105	Notice Stage LED is blinking
[101] [Thru] [105] [Out]	The changes are stored immediately to Cue 105, because we are in standard Blind.
[Live]	Back to what was being edited
[Stage]	Adds the Staging Buffer back onto Live
[102] [+] [104] [Enter], sharpen, Make them cyan	All the data in the table display is in orange
[Select Last] {All NPs} [Copy To] [Sneak] [5] [Enter]	Moves fixtures into position without intensity
[Clear], then [Copy To] [Sneak] [3] [Enter]	Clear the channel selection, then fade the intensities into Live
[Stage] [Stage]	Clear out of Staging Mode
[Update] [Cue Only] [Enter]	Update the changes Cue Only to Cue 102.5



Hierarchy & Inspector Tools

In the Augment3d Edit area, the Hierarchy menu provides tools for selecting and organizing the objects in your model.

The Inspector menu, which opens when something is selected in the Hierarchy, allows you to make property modifications to the objects.

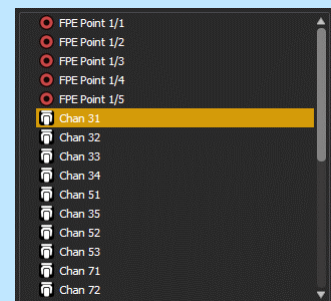
HIERARCHY

MANAGING OBJECTS

In the Augment3d window, click on {Edit}



Drag and re-arrange objects into whatever order you like (be careful not to nest objects under other objects)



Use the {Sort} options: Unsorted (User Sort - default), Sort A > Z, Sort Z > A



GROUPS (FORMERLY KNOWN AS FOLDERS)

Click the hamburger menu, and click on {Add Group}

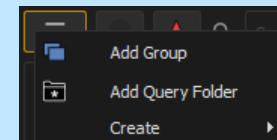
Find the New Group at the bottom of the list

Double click it, and rename it "Moving Lights"

To select multiple objects, hold [Control] on the keyboard, and click objects you want to select

Using [Control], select Channels 111 & 112, and drag them onto the Group

Click the arrow next to the {Moving Lights} group, you will see the channels inside that Group



RENAMING OBJECTS

Double click on {Truss Square}, and rename it to "FOH Truss"

It is possible to rename channels so be careful doing that. Once you name something, it may be difficult to find that channel number again.

NESTED OBJECTS

Using [Control], select channels 1 – 10 & 101 – 105, and drag them onto the {FOH Truss}

Click the arrow next to the {FOH Truss} object to expand it

You will see the channels nested under that object

Zoom the camera out so you can see the Truss

Click on the Truss object

Everything nested under the Truss object also gets selected

Notice when things are selected in the model, they are automatically selected in the Hierarchy as well.



TO CHOOSE A NESTED OBJECT

Use the Hierarchy – click on {Channel 101} under the {FOH Truss} object

Just 105 gets selected

Hold [Shift] and click on {Channel 105}

All 5 channels are selected, like Thru

For now, just select {Channel 105}

INSPECTOR

Click on {FOH Truss} to look at the Inspector

OVERVIEW

The Inspector lets you change object properties.

- **Name** – You can change the name of the object. Be careful changing channel names, as there is no link to the channel number if you relabel it.
- **Icon Color** – This will change the color of the object's icon in the Hierarchy.
- **Lock** – If an object is locked, you cannot make any changes to it.
- **Visibility** – shows or hides the object in the Augment3d model.
 - If objects are associated with other objects so as the FOH Truss, if you hide the truss, all those objects are hidden as well
- **Quick Rotate Buttons** – Rotate the object in 90-degree increments around the X, Y, or Z axis.

Add object from Library > Stock > Furniture > Chair Desk

Move the camera in to see the chair and what happens

Click on the chair and use the Quick Rotate buttons for X, Y, and Z

- **Move Object to the Ground Plane** – sets the bottom edge of the object at Z=0.

In the field {Position Z}, make the height be 5'-0"

Click Move Object to the Ground Plane

- **Tags** – Useful in Augment3d search. Some tags are auto-generated (such as with channels) or pulled from the Patch Database fields. Other tags can be added in Augment3d using the **Add Tag** field below.

In the Hierarchy, Select {Channel 122}

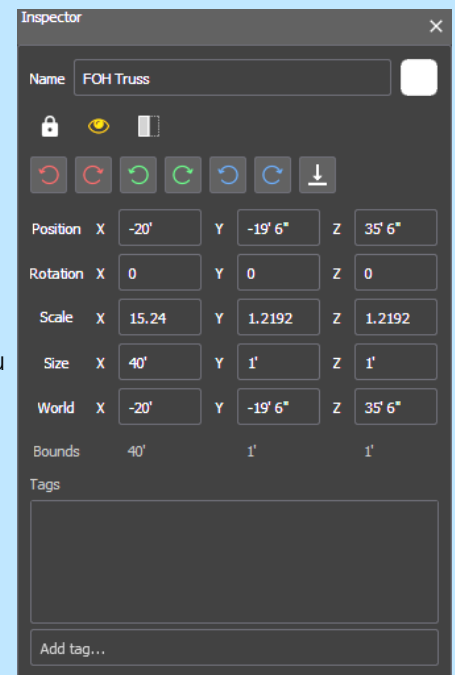
Using [Control], select channels 1 – 10

Click in the {Add a Tag} field, and type FOH, then [Enter]

In Hierarchy at the top, in the Search field, type FOH

Click on the {X} in the tag

Click {Done} and {Apply Augment3d Model Changes}



Chair will be up in the air

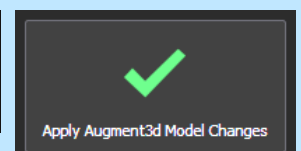
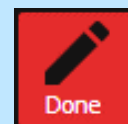
Snap the chair to the ground plane

Notice some tags exist already

Channels now have the tag FOH

Hierarchy repopulates to only items with a FOH in its tag – truss, movers and channels 1 - 10

This removes the tag from the channels



BEST PRACTICE IDEAS

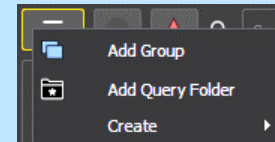
- Using {Lock} on scenic objects once they have been placed will prevent them from accidentally being selected and moved.



QUERY FOLDER

A Query Folder will automatically contain objects that meet certain criteria that you define.

In the Hierarchy, click the hamburger menu, and click on {Add Query Folder}



Find the Query Folder at the bottom of the list

OVERVIEW

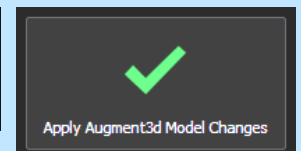
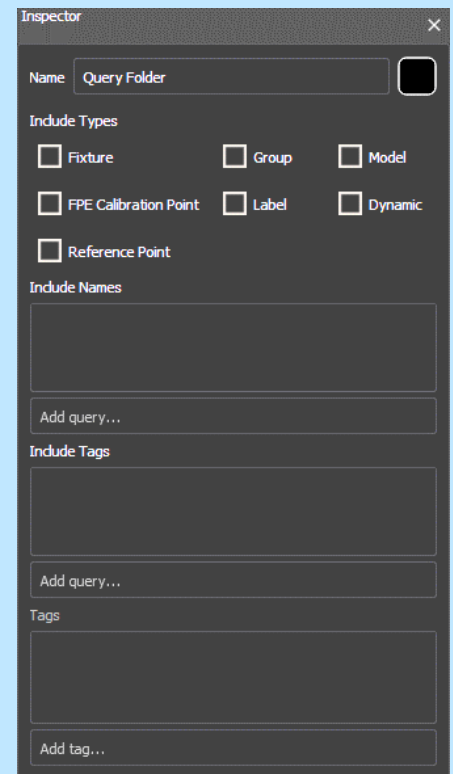
The Inspector contains the settings for the Query Folder:

- **Name** – You can change the name of the Query folder.
- **Icon Color** – This will change the color of the icon in the Hierarchy.
- **Include Types** – What types of objects do you want to be able to be included in the folder? With no Included Names or Included Tags defined, all objects of that type will appear in the folder.
- **Include Names** – Within the Included Types, objects with exact name matches will appear.
 - For example, if Dynamic is not checked in the area above, then typing FOH Truss in Include Names won't find the truss object.
- **Include Tags** – Within the Included Types, objects with exact tag matches will appear.
- **Tags** – These are the tags for the Query Folder, used just like tags on other objects. They do not have to do with the Query itself.

Wildcards for Included Names & Included Tags:

- **Asterisk (*)** – allows for wildcard. “*MTG*” will return all Included Types that have “MTG” anywhere in the name.
- **Question Mark (?)** – This will allow wildcards for the character that the question mark is occupying. For example, “Chan ??” will return all channels that have double-digit numbers. “Chan ???” would return triple-digit channels.

Click on {Done}, and {Apply Augment3d Model Changes}



BEST PRACTICE IDEAS

- How much time do you really want to spend in this area? Much like Keywords in Patch – how useful will the organization be compared to the time you put in?



Scenic Element Channels

Scenic Elements don't always stay in one place. Discover how Eos channels can be tied to Augment3d elements to change position, rotation and visibility on the fly as well as in cues.

PATCHING & NESTING SCENIC ELEMENT CHANNELS

[Go To Cue] [Out] [Enter]

[Address] [Address] to get into Patch

[1001] {Type} ETC Fixtures > Scenic Element > Moveable [Enter]

There are two profiles of Scenic Elements:

- **Scenic Element** - can be given intensity values to show or hide them on a cue-by-cue basis.
- **Moveable** – can be given intensity values, but also have position, orientation, and rotation changed from cue-to-cue.

[Live], [1001] [Enter]

[Full] [Enter]

In the Augment3d window, {Edit}

Double click {Chan 1001} in the Hierarchy, and relabel it "CH 1001 – Truss Movable" [Enter]

Drag & Drop {FOH Truss} onto {CH 1001 – Truss Movable}

Notice the two levels of nesting.

In the Hierarchy, select {Channel 101}.

The Position XYZ fields are Channel 101's location in relation to the base point of its Parent Object (The Truss – remember the base point is at the far end)

The World XYZ fields are Channel 101's location in relation to the Augment3d origin.

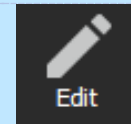
When we nested the objects, they did not move their World location, and adjusted their Position location relative to their parent.

Click on {Done}, and {Apply Augment3d Model Changes}

Patch a movable scenic channel, which will be used to control model objects.

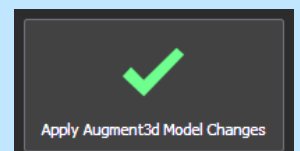
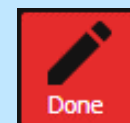
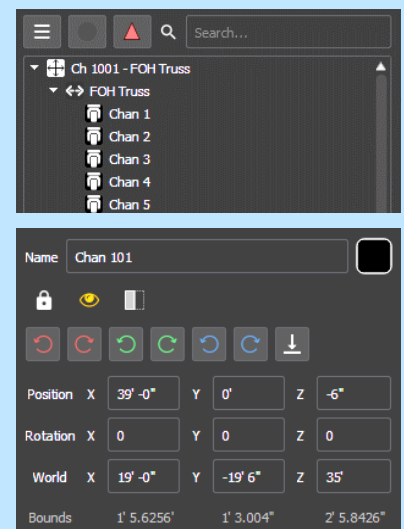
Scenic Element Channels use intensity for visibility, and they home to "Hidden"

Ensures we see the models we added to the channel in Augment3d Edit Mode



Rename the object

Nests the Truss object and the channels under Channel 1001



CONTROLLING & RECORDING SCENIC ELEMENT CHANNELS

FOH TRUSS

In Live	The channel is still at Full, so we can see the nested model objects.
[1001] [Out]	Truss and all the fixtures are no longer visible.
[Full] [Full]	Allows us to see them again.
[1001] {Z Focus} [At] [+] [-] [5.6] Enter]	Truss lowers by 5'-6"
[Record] [202] [Enter]	Stores the lower position in Cue 202.
[1001] {Z Focus} [At] [+] [10] [Enter]	Truss raises by 10'-0"
[Record] [203] [Enter]	Stores the higher position in Cue 203.
[■] (Back)	Watch the Truss move to its position stored in Cue 202.
[▶] (Go)	Watch the Truss move to its position in Cue 203 in the cue time.

CHAIR

[Address] [Address] to get into Patch	
[1002] [Type] ETC Fixtures > Scenic Element > Moveable [Enter]	Patches another Scenic Element Channel
[Live]	
In Augment3d window, {Edit}	This creates a folder containing the truss (and all channels still assigned to the truss)
In the Hierarchy – drag and drop {Chair Desk} onto {Chan 1002}	The chair disappears. Channel 1002's intensity is still at 0.
{Done} editing, { Apply Augment3d Model Changes}	Exit Augment3d Edit Mode
[1002] [Full] [Full]	The chair becomes visible.
On the [Focus] XYZ Encoder Page, use the encoders to move to another position. Use the encoders to rotate, and notice the fulcrum of the rotation.	
[Update] {Cue Only} [Enter]	Updates the chair visibility and location into Cue 203.
[1002] [Enter], put it in a different position	
[Record] [204] [Enter]	Records the chair into a new position.
[■] (Back)	Moves the chair to its position in Cue 203.
[▶] (Go)	Move the chair to its position in Cue 204, in the cue time.

BEST PRACTICE IDEAS

- Record Presets with scenic setups for each scene, and store in a second cue list.



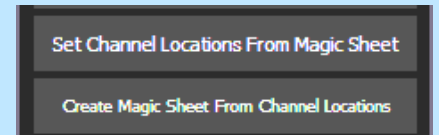
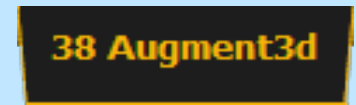
Augment3d and Magic Sheets

We spend a lot of time placing fixtures in space so why not leverage that work by creating a magic sheet that is representative of lights in the rig.

For these features, you will need access to an Augment3d tab.

On a console or ETCnomad running Augment3d in a tab, simply double click on the Augment3d tab to see the tools.

With a tether, you need to open an Augment3d tab on the connected console, and then double click on the Augment3d tab



AUGMENT3D ↔ MAGIC SHEETS

You can move your channel positions from your Augment3d model into a starter Magic Sheet, or vice versa.

Add an Augment3d tab, double click on the tab

Click on "Create Magic Sheet from Channel Locations"

Notice several angle options, including current camera view.

Select {Top-Down}, use Magic Sheet #401, and click {Confirm}

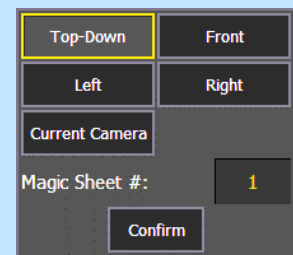
Label it "Aug3d Magic Sheet", and hit [Enter]

Click on Add-A-Tab, and open a Magic Sheet (3) tab.

Use the MS Browser to select Magic Sheet 401

This is a starting place, you can edit it to make it your own.

Don't forget you can use the "Change Type" button if you don't like the symbol used.



As soon as hit Enter, it is created.

AUGMENT3D FULL SCREEN (LEM)

Like Magic Sheets, Augment3d tabs on a console or ETCnomad can be opened into a full-screen mode or **Limited Expand Mode**.

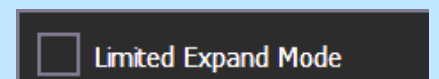
Click on the gear to the left of the Augmented tab

Click "Limited Expanded Mode"

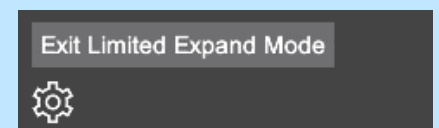
Notice in Limited expand Mode, there is still a gear in the lower left corner.

Click on that gear

Click on "Exit Limited Expanded Mode."



Augment3d fills the entire window



BEST PRACTICE IDEAS

- There is also a Set Channels from a Magic Sheet. If you already have a Magic Sheet with all your channels in a plot format, you can do this, but it will take a lot of time to then position the fixtures in the Augment3d model.



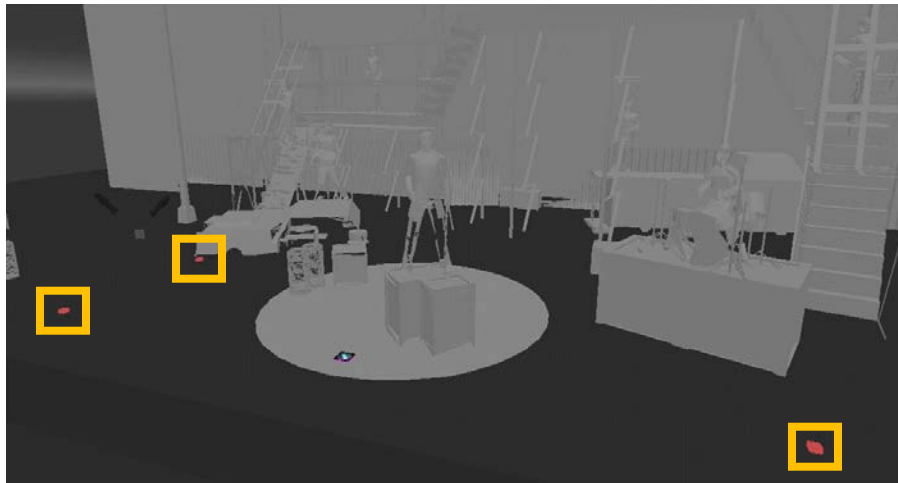
Appendix 1: FPE Tool

There are three ways to add fixtures to an Augment3d model: import from Vectorworks, manually add all the data (Patch and Position) or use Augment3d's FPE feature, or Fixture Position Estimation. It automatically determines the location of your automated fixtures. To start, you need a real rig, and a few focus palettes, and then let Augment3d calculate the position of the fixtures.

QUICK START

To have FPE place fixtures in 3D space in the model, you will need:

- 4 to 8 focus palettes at precisely known locations. The further apart they are, the more accurate FPE will be.
- Reference points in your Augment3d model in the exact locations as the focus palettes, set to the same focus palette number.



In the Augment3d tab, click on {Edit}

Every time you start a new show file, you are given 4 FPE points, They will show up in the model and in the Hierarchy. You will want to put those in the same locations in both the model and in the venue/space. All calculations will be off those points. This means measuring and marking the locations on the stage and then placing them in the model at the same measurements.

Select the fixtures and bring to Full (102 + 104 + 111 + 112)

Iris or zoom down to smallest beam

To center beam when focusing

Make the edge as sharp as possible

To make as precise as possible

Focus all of the lights to the predetermined locations.

Make sure the center of the beams are at the exact same location.

A crosshair gobo may help

Record the Focus Palette. Repeat for at least 4 Focus Palettes.

More FPE palettes add accuracy, but there is a maximum of 8 that can be used.



Go into Patch > {Augment3d} > {FPE}

Select the channels on the command line

In the Augment3d section, click on the FPE tab

To the right of the Position & Orientation Tab

Position & Orientation		FPE			
FPE Set	Label	Point	FP #	(X, Y, Z)	
FPE Set 1		1	901	(-15' 6", 0', 0')	
		2	902	(-15' 6", 8' 8", 0')	
		3	903	(15' 6", 8' 8", 0')	
		4	904	(15' 6", 0', 0')	
		5	905	(0', 5' 2", 0')	
Add FPE Set Delete FPE Set		Add FPE Point Delete FPE Point			

You can add or delete FPE points as necessary. If you have already associated the FPE points with the Focus Palettes in the model, they will show up in the FP # column, with the XYZ position of the FPE point.

You have the ability to organize FPE Points and Focus Palettes into FPE Sets. You can create different groupings of FPE points so that if you have fixtures that cannot all reach the same four points or if you are working in multiple spaces you can use the groupings to recalculate your positions for different fixtures.

Click or touch on {Add FPE Set}.

To get a popup to make the change

Touch the Label field

To label the set

Click or touch on {Delete FPE Set} then again on {Confirm Delete}

To remove a set

To add an FPE point:

Double click or double touch on the (X,Y,Z) Field.

To get a popup to make the change

In Patch, use the {Add FPE Point} button

To add additional FPE points

A little red dot will show up in the model at 0/0/0. You can reposition either in the model or in the Patch – FPE tab.

Click or touch on {Delete FPE Point} then again on {Confirm Delete}

To remove a FPE

By default, when we give you the four default FPE points, we assign default focus palettes 901 through 904 to them.

To associate different Focus Palettes with the FPE point:

Double click or double touch on the FP# Field.

To get a popup to make the change

Keep in mind that all this information is based on channel selection and you can get errors if you have channels that are not part of your FPE set or your individual focus palettes.

[101] [Thru] [105] [Enter]

To select different channels

Use the {Recalculate FPE} button and hit {Confirm FPE}

Watch the button to see the recalculations for each channel

Appendix 2: Focus Wand

The Eos remote app now includes a feature called Focus Wand, which leverages the device's Augmented Reality kit to bridge between reality and Augment3d. Some features include:

- Augment3d knowing the device's position relative to the model and allowing the device to point moving fixtures like a laser pointer, or to point the fixtures at the device in its current location.
- Showing channel boxes and labels overlaid onto the device's camera feed.
- Allowing for selection through the device's camera feed, as well as some parameter control, like intensity and zoom.
- Displaying channel labels where a beam intersects with objects.



GETTING READY – CREATING AR TARGETS

The phone or tablet device needs to orient itself relative to the model. To accomplish this, an AR target is placed in the exact same position in the model. Once the camera recognizes the AR target with its camera, it uses its internal sensors to track the device's position and orientation, relative to the AR target. From this, Augment3d can use that position for its features.

In the Augment3d window, click {Edit}.

In the Hierarchy, click on the hamburger icon

Click {Create} and then click {Reference Point}

Reference points have a special property, Image. Image allows you to pick one of the provided AR targets and put that in the model.

In the Inspector, click on {Image}

Click on {Choose Image}

Select a Reference Point image – Reference Point 1

There are 10 reference point images available. You can use each one once anywhere in your space. You can use multiple reference points to increase accuracy.

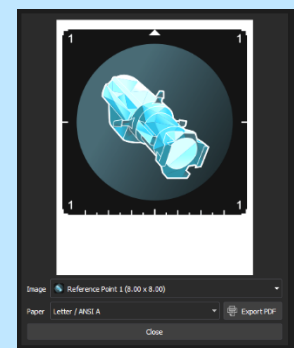
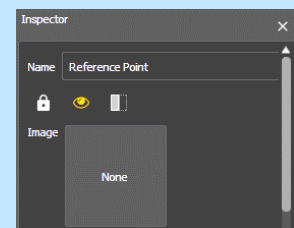
You can also choose what size paper it will be on to print.

Click {Okay}

Make sure that this image is in the exact place and orientation where you place a reference point on stage. If the target in the model and the one in reality are mismatched, the Focus Wand will be inaccurate.

Use Click to Move and Click to Rotate

Click {Done}, {Apply Augment3d Model Changes} to close the editor.



The image appear on the model

To put target in exactly the same place and orientation as it will be in reality

Let's place another target

In the Augment3d window, click {Edit}.

In the Hierarchy, click on the hamburger icon

Click {Create} and then click {Reference Point}

Remember you can change the labels in the Hierarchy by double clicking the object.

In the Inspector, click on {Image} , Click on {Choose Image}

Select a Reference Point image – Reference Point 7

Click {Okay}

Use Click to Move and Click to Rotate

Position in the exact same XYZ position as it will be in reality

Click {Done}, {Apply Augment3d Model Changes} to close the editor.

PRINT REFERENCE POINT IMAGES

To print the images, you'll need a thumb drive/USB drive. The images are PDF files.

[Displays], File > Export > Augment3d Reference Point Images

Select your destination drive and hit [Select]

The same dialog box as is in the Inspector appears

Select the Reference Point images needed, and the paper size you will be using to print.

Click {Export PDF} to save the file to the chosen location.

Dialog box remains open

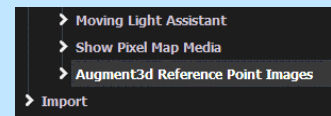
Repeat with as many targets as you want

then click {Close} to leave the export window

Print the target(s) from your printer

It is very important to print them on the same sized paper that was specified in the model and without any scaling. The Focus Wand device's AR kit will use the size of the image as part of its calculations – so precision is key.

Place the AR targets on the stage in the exact same location as they are in the model, paying attention to orientation.



Appendix 3: Vectorworks Plugin

There are ETC-developed plug-ins for Vectorworks 2019, 2020, and 2021 – which allow you to export fixture data, and easily import it into Eos Patch.

INSTALLING THE PLUG-IN

To install the plug-in, download it from our website and follow the instructions:

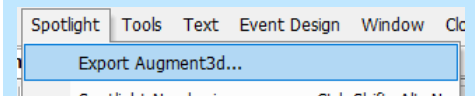
[Augment3d Vectorworks Plug-in v1.0](#)

EXPORTING FIXTURE DATA FROM VECTORWORKS

Once the plug-in is installed, it is simple to export a file.

In Vectorworks, go to {Spotlight}, {Export Augment3d...}

You will be prompted to save a *.a3d file. Choose a name and location, and click {Save}

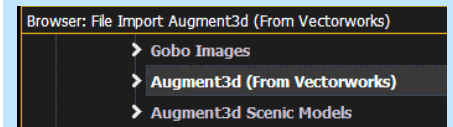


All objects that are designated as Fixtures in Vectorworks Spotlight will have their database information exported.

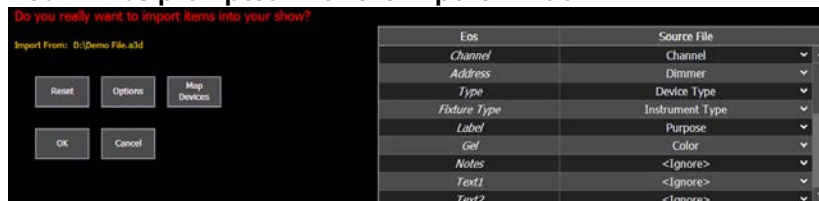
IMPORT *.A3D FILE INTO EOS

In Eos, Go to Browser > File > Import > Augment3d (From Vectorworks).

Choose the file, and hit [Select] or [Enter]

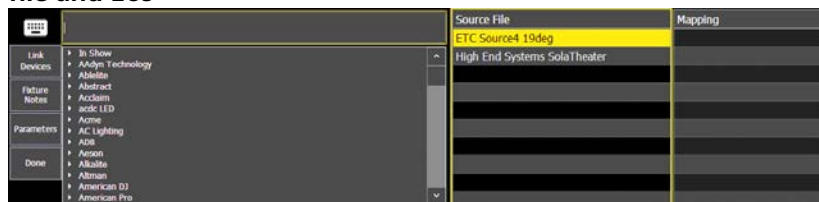


You will be prompted with the Import window:

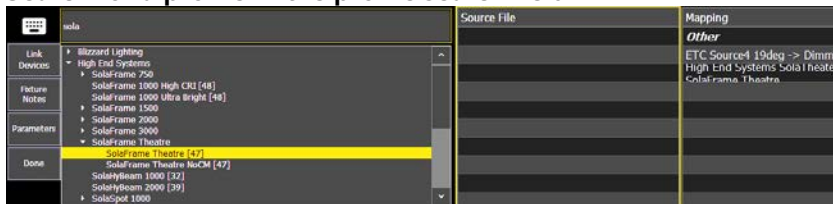


On the right, you can match Eos patch fields with those found in the *.a3d file. Eos makes default recommendations

Click on {Map Devices} to link fixture profiles between the *.a3d file and Eos



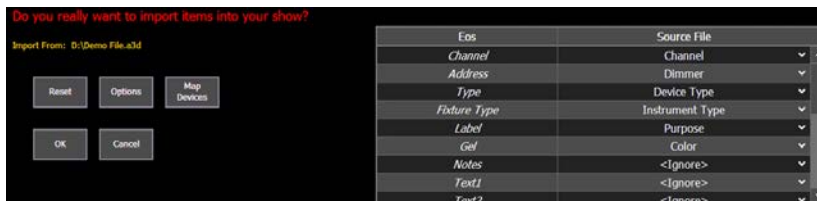
Search for a profile in the profile search field:



Once selected, select the cooresponding Source File profile, and click {Link Devices}. The link will now show up in the Mapping column

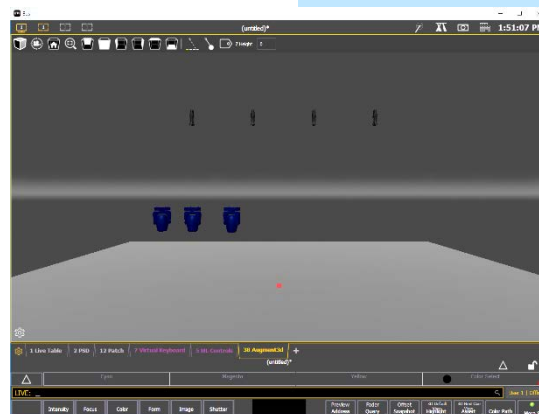
Continue until all Source File profiles are linked to Eos profiles

Click {Done} to return to the main menu:



Click {OK} to finish the import

Fixtures will now show up in Patch and in the Augment3d window



Appendix 4: Vectorworks Model Export

You can export your non-fixture model elements from Vectorworks and import them into Eos as model objects.

PREPARING VECTORWORKS FOR EXPORT

Vectorworks' export engine relies on model visibility states. When you export, only objects that are visible will be exported, and all graphical and pose properties of that object will be retained (with certain limitations provided by the file type).

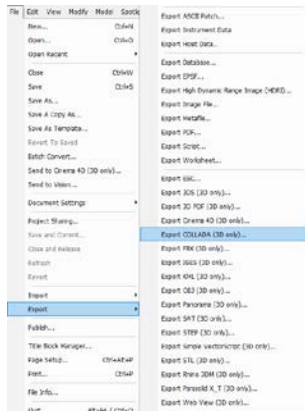
A few things to consider before exporting model objects from Vectorworks:

- Complexity – a lot of models have extreme detail, because they were created for engineering or building purposes (for example, some have been known to include bolts with threading – a detail you likely do not need for lighting the object). Consider what level of detail will be useful in Augment3d and adjust your model accordingly. Remember that every additional surface/triangle adds processing overhead to the device hosting Augment3d.
- Visibility & Movability – Augment3d has the ability to show & hide objects, as well as attach them to a channel for movement. Before exporting, consider what separate components you want to control, and export them individually. For example, exporting the architecture, separate from masking, separate from moving scenic elements may be more useful than one big model export including them all.
- Location – many file formats support remembering an object's location and orientation, relative to Vectorworks' origin point. This can be useful when you are exporting and subsequently importing objects that need to maintain relative location. For example, if you want to import a theater's architecture, and then import individual linesets in their proper location – using a file format that supports origin makes the process a lot easier.

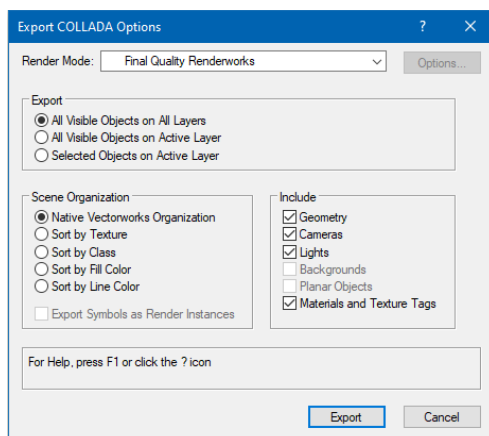
EXPORT COLLADA (*.DAE) FILES

The Collada file type is preferred, because it supports many of the properties that make transferring 3D objects between programs easy.

To export a Collada file from Vectorworks, set your visibility states via Layers or Classes to the desired export:



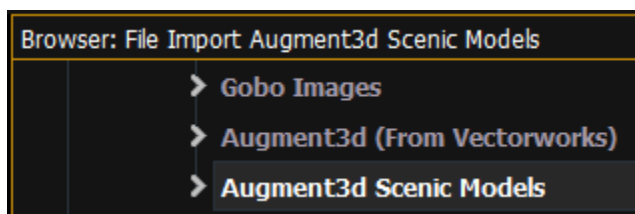
Once the save dialog shows up, use Vectorworks' default settings:



Click {Export}, and then save the file in your preferred location.

IMPORT COLLADA (*.DAE) FILES INTO EOS

We cover importing model files in this workbook, in the chapter titled “Importing 3D Objects.” Remember that importing files with Augment3d on a console is different than if Augment3d is running as a tether.





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